

UNIT - III

3

IoT : Design Methodology

Syllabus

Steps, Basics of IoT Networking, Networking Components, Internet Structure, Connectivity Technologies, IoT Communication Models and IoT Communication APIs, Sensor Networks, Four pillars of IoT: M2M, SCADA, WSN, RFID.

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3.1 IoT Design Methodology Steps

➡ **SPPU : May-18, Dec.-18, May-19, Dec.-19, March-20**

- The rise of the Internet of Things has led to an explosion of new sensor computing plat-forms. The complexity and application domains of IoT devices range from simple self-monitoring devices in vending machines to complex interactive devices with artificial intelligence in smart vehicles and drones.
- As IoT developers wish to meet more aggressive platform objectives and protect market share through feature differentiation, they must choose between low-cost, and low-performance CPU-based Commercial-Off-The-Shelf (COTS) systems, and high-performance custom platforms with hardware accelerators such as GPU and FPGA.
- An IoT platform facilitates communication, data flow, device management, and the functionality of applications. The goal is to build IoT applications within an IoT platform framework. The IoT platform allows applications to connect machines, devices, applications, and people to data and control centers.
- Home automation can be described as introduction of technology within the home environment to provide convenience, comfort, security and energy efficiency to its occupants.
- A home automation system can involve switching off electrical appliances like air-conditioners or refrigerators when a desired temperature has been reached, then switching on again when the temperature has crossed a certain value
- Fig. 3.1.1 shows design methodology steps.
- **Sensors :** Sensors are the eyes of a home automation system. They "see" the environment and convert what they find into an electrical quantity that can be measured by a microcontroller or system processor.
- **Remote Connectivity :** Depending on need and various design considerations, users may need to be able to control the system and appliances remotely.

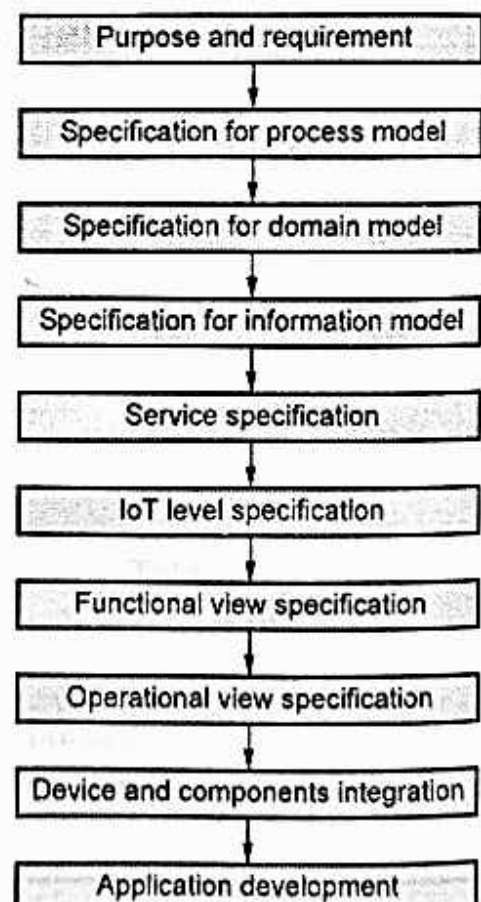


Fig. 3.1.1 : Design methodology steps

3.1.1 Purpose and Requirement Specification

- Purpose : Using web application, user can remotely control the home electronic devices (light and air conditioner)
- Behavior : Home automation system will work in two modes : manual and automatic mode. In manual mode, web system provides options of manually and remotely switching ON / OFF the light. In automatic mode, sensor is provided in the room. System measures the darkness of the room and light is ON when darkness increases. Room temperature is measured by another sensor, air conditioner is ON when room temperature is increases.
- System Management Requirement : Remote monitoring and control function is provided by system.
- Data analysis requirement : Data analysis is based on local data.
- Application deployment Requirement : Application is installed on local device but it support remote operation.
- Security requirement : Basic authentication mechanism must be provided.

3.1.2 Process Specification

- Process specification is defined after requirement step. Draw the use case based upon first stage. Fig 3.1.2 shows process specification diagram.

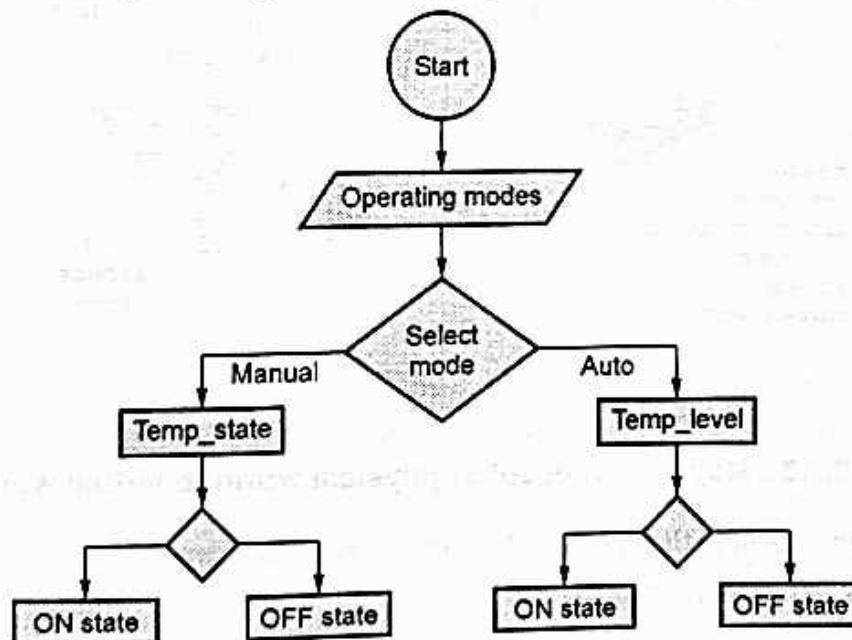


Fig. 3.1.2 : Process specification diagram

- In auto mode, system monitors the room temperature and light level and takes the decision for switching ON / OFF.
- In manual mode, user performs the operation.

3.1.3 Domain Model Specification

- A domain model defines the main concepts of a specific area of interest.
- The IoT is a support infrastructure for enabling objects and places in the physical world to have a corresponding representation in the digital world. Fig 3.1.3 shows mapping concept of physical world to virtual world.
- As interaction with the physical world is the key for the IoT; it needs to be captured in the Domain Model (DM). The DM defines the main concepts of the Internet of Things and the relations between these concepts.
- User and a physical entity are two concepts that belong to the domain model. A User can be a human user, and the interaction can be physical.

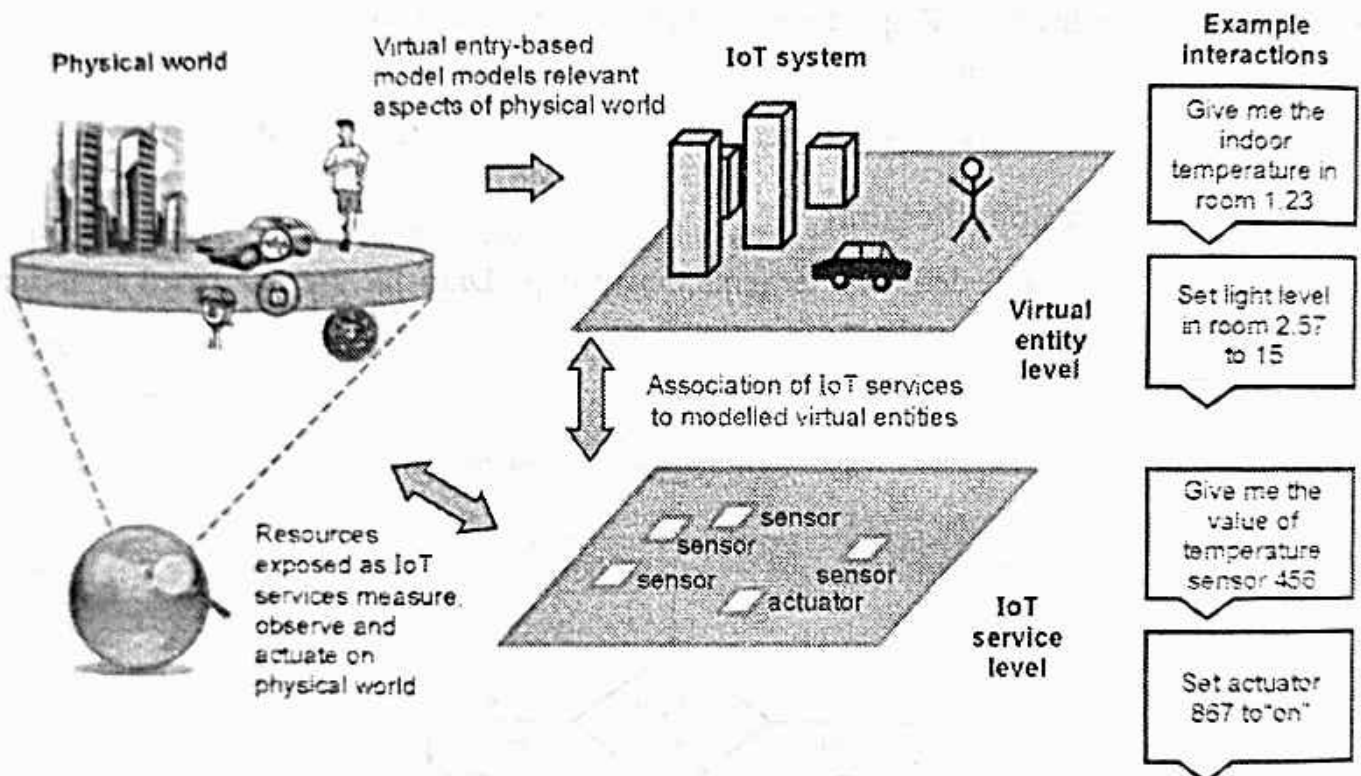


Fig. 3.1.3 : Mapping concept of physical world to virtual world

- The physical interaction is the result of the intention of the human to achieve a certain goal. Fig. 3.1.4 shows IoT domain model.
- A Physical entity, as the model shows, can potentially contain other physical entities; for example, a building is made up of several floors, and each floor has several rooms.
- A Physical entity is represented in the digital world as a Virtual entity. A Virtual entity can be a database entry, a geographical model, an image or avatar, or any other Digital Artifact.

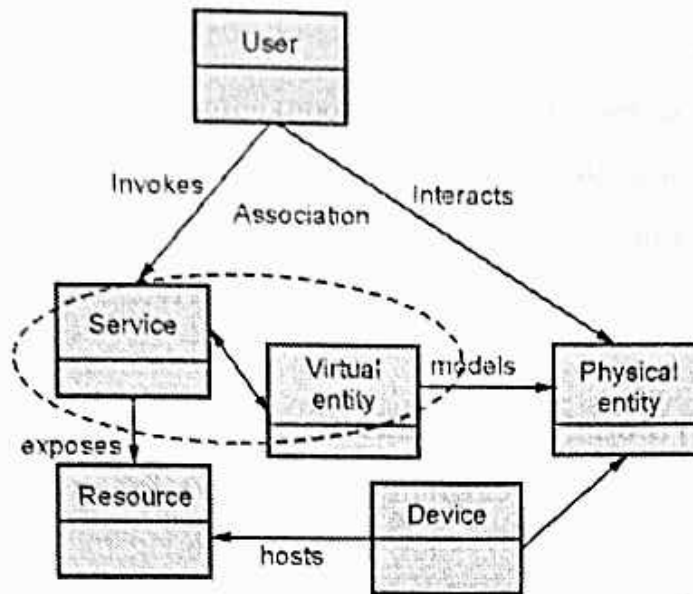


Fig. 3.1.4 : IoT Domain model

- The relations between services and entities are modeled as associations. These associations could be static, e.g. in case the device is embedded into the entity; they could also be dynamic, e.g., if a device from the environment is monitoring a mobile entity. These identified concepts of the IoT domain and the relations between them are depicted in Fig. 3.1.5.

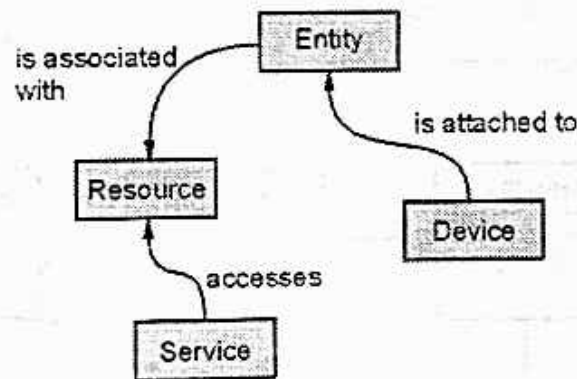


Fig. 3.1.5 : Key concepts and interaction in IoT model

- One physical entity can be represented by multiple virtual entities, each serving a different purpose. For the IoT domain Model, three kinds of device types are the most important :
 - Sensors** : These are simple or complex devices and contain a transducer that converts physical properties such as temperature into electrical signals. These devices include the necessary conversion of analog electrical signals into digital signals.
 - Actuators** : These devices that involve a transducer that converts electrical signals to a change in a physical property.
 - Tags** : Tags in general identify the Physical entity that they are attached to.

- Home Automation System Example :

1. **Physical entity** : Room in the home and room temperature
2. **Device** : Single board mini computer with sensor and relay switch
3. **Resources** : Operating system which runs on mini computer
4. **Services** : Mode selection, controller service which runs services on the device, retrieve the room temp.

3.1.4 Information Model Specification

- An abstract description (UML diagram or ontology) for explaining information about elements or concepts defined in the IoT Domain Model.
- The information model models domain model concepts that are to be explicitly represented and manipulated in the digital world. In addition the information model explicitly models relations between these concepts.
- Fig. 3.1.6 shows information model. The information model is a meta model that provides a structure for the information. This structure provides the basis for defining the functional interfaces.

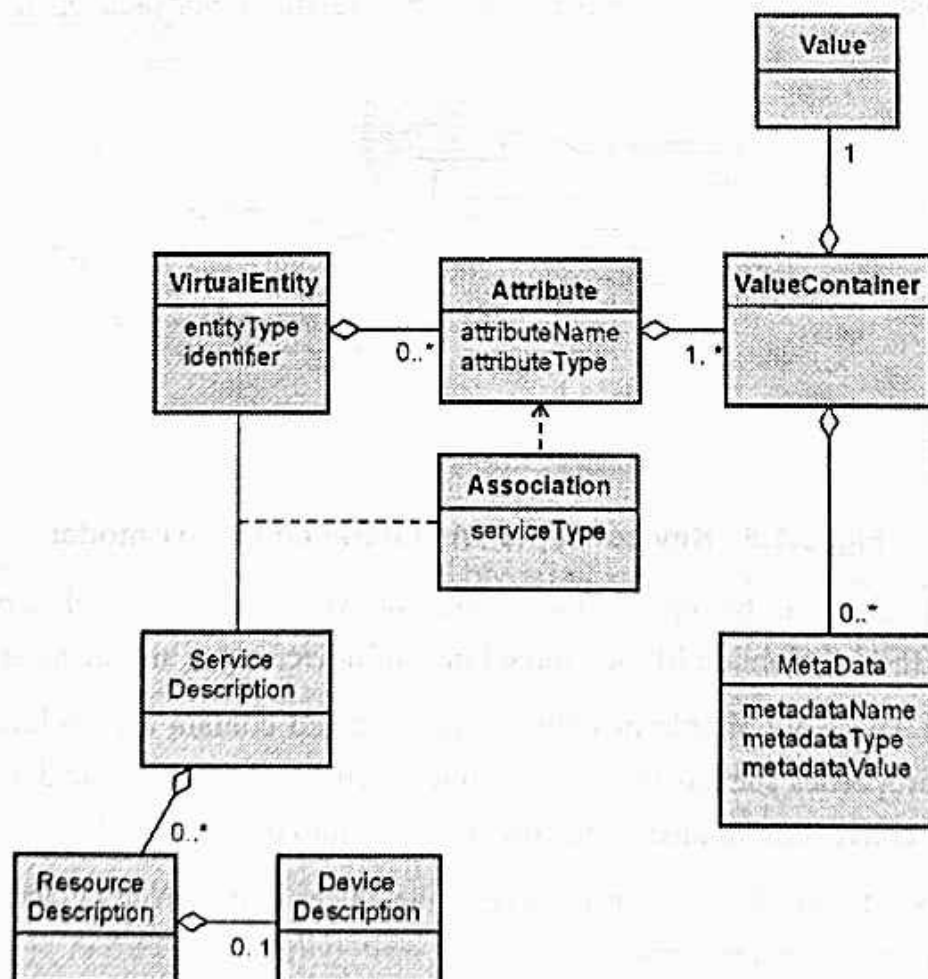


Fig. 3.1.6 : IoT Information Model

- IoT Information Model is represented using Unified Modeling Language (UML) diagram. The IoT Information Model maintains the necessary information about Virtual Entities and their properties or attributes.
- The information model for an object can contain information about the objects structure and resource types. This can enable APIs to automatically be composed by middleware and automatically consumed by application software.
- Additional metadata can indicate context, such as geographical location, and bindings, such as message protocols and event handlers, as well as access control information.
- The IoT Information Model describes Virtual Entities and their attributes that have one or more values annotated with meta-information or metadata. The attribute values are updated as a result of the associated services to a Virtual Entity.



3.1.5 Service Specification

- Service specification in IoT defines list of services provided by the system. It includes service type, input / output, service endpoints, schedulers, service effects and service preconditions.
- State and attributes are defined in process specification and information model. For example in home automation system, mode setting is an example of mode service. User can set the auto or manual mode.
- Mode services sets mode to auto or manual or retrieves the current mode.
- In manual mode, the controller service, retrieves the current state from the database and switches the air conditioner ON / OFF.

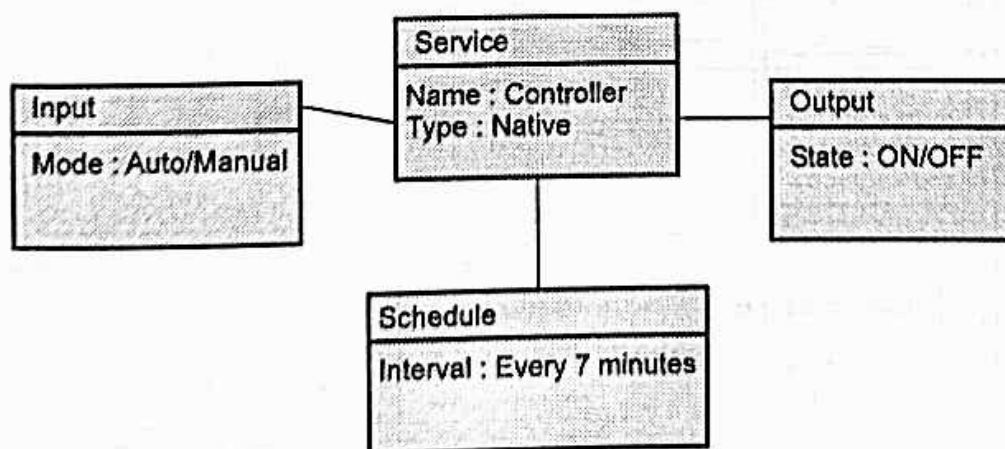


Fig. 3.1.7 : Home automation Controller service

3.1.6 IoT Level Specification

- IoT Development level are six types.

IoT Level 1	Single node, perform sensing, perform analysis and hosts the application
IoT Level 2	Single node, perform sensing, perform local analysis
IoT Level 3	Single node, data is analyzed in cloud and application is cloud based
IoT Level 4	Multiple Node, perform local analysis, application is cloud based
IoT Level 5	Multiple end node and coordinator node,
IoT Level 6	Multiple independent node, perform sensing, send data to cloud

3.1.7 Functional View Specification

- Functional View describes the system's runtime Functional Components, their responsibilities, default functions, interfaces and primary interactions. The Functional View derives from the Functional Model and reflects the developer's perspectives on the system.
- It will need to be extended with all identified (and recommended) new profile-specific Functional Components including their interfaces and a list of Sequence Charts illustrating recommended usage of those components.
- Fig. 3.1.8 shows functional view.

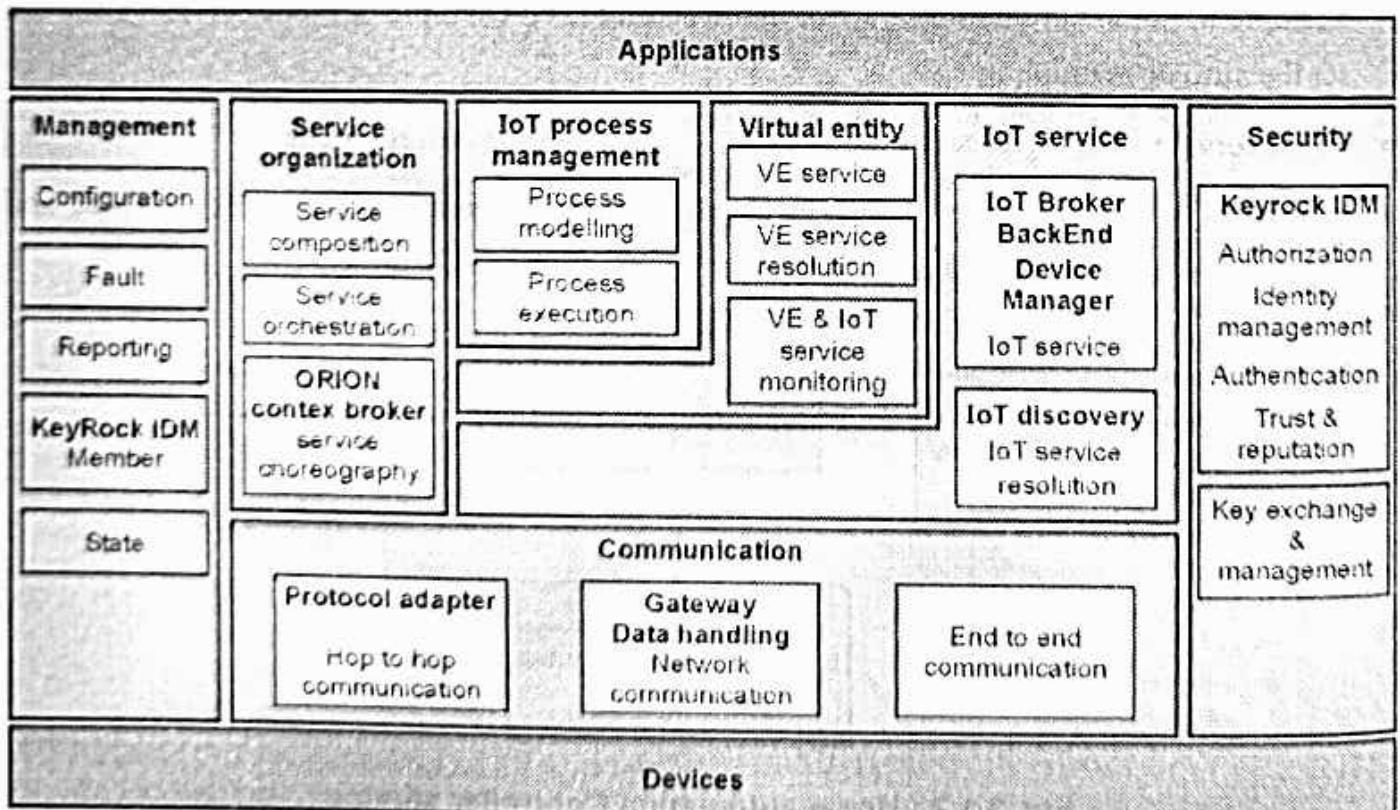


Fig. 3.1.8 : Functional view

- The viewpoints used for constructing the IoT Functional View are hence :
 - 1) The Unified Requirements;
 - 2) The IoT Functional Model
- Once all Functional Components are defined, the default function set, system use cases, sequence charts and interface definitions are made.
- **Device and Application functional group** : Device functional components contains the sensing, actuation tag, processing and storage components. Application functional group contains standalone application.
- **Communication functional group** : It contains the components for end-to-end communication, network communication, and Hop-by-Hop communication.

Communication type	Description
End-to-End Communication FC	1) Responsible for end-to-end transport of application layer messages through diverse network and MAC/physical layers 2) Used with mesh radio networking technologies such as IEEE 802.15.4 3) The End-to-End FC interfaces the Network FC on the "southbound" direction.
Hop-by-hop FC	1) Responsible for transmission and reception of physical and MAC layer frames to/from other devices. 2) Two interfaces used : one "southbound" to/from the actual radio on the device, and another for "northbound" to/from the Network FC in the Communication FG.
Network FC	1) Responsible for message routing and forwarding and the necessary translations of various identifiers and addresses.

- **IoT Service Functional Group** : It consists of IoT Service FC and the IoT Service Resolution FC. Various service implementations are covered in service FC and service resolution FC contains the necessary functions to realize a directory of IoT Services that allows dynamic management of IoT service descriptions.
- **Virtual Entity Functional Group** : The Virtual Entity FG contains functions that support the interactions between Users and Physical Things through Virtual Entity services.
- **Process Management FG** : Provides the functional concepts necessary to conceptually integrate the IoT world into traditional (business) processes.

- The Process Modeling FC which provides the tools required for modeling IoT-aware business processes that will be serialized and executed in the Process Execution FC, which is responsible for deploying process models to the execution environments.
- **Service Organization FG** : Acts as a communication hub between several other Functional Groups by composing and orchestrating Services of different levels of abstraction.
- The Service Orchestration FC resolves the IoT Services that are suitable to fulfill service requests coming from the Process Execution FC or from Users while the Service Composition FC is responsible for creating services with extended functionality by composing IoT services with other services.
- Service Choreography FC offers a broker that handles Publish/Subscribe communication between services.
- **Virtual Entity FG** : Provides functionality for the interaction of VEs with the IoT system, for VE look-up and discovery and for providing information concerning VEs. The VE Resolution FC provides discovery services for associations between VEs and IoT.
- VE and IoT Service Monitoring FC is responsible for automatically finding new associations based on service descriptions and information about VE's. the VE Service FC handles entity services.
- **Service FG** : Provides IoT services as well as functionalities for discovery, look-up, and name resolution of IoT Services.
- **Security FG** : It is responsible for security and privacy matters in IoT-A-compliant IoT systems.
 - 1) The Authorization FC is used to apply access control and access policy management while, the Authentication FC is used for user and service authentication.
 - 2) Key Exchange and Management (KEM) FC enables secure communications ensuring integrity and confidentiality by distributing keys upon request in a secure way.
- **Management FG** : It is responsible for the composition and tracking of actions that involve the other FGs.
 - 1) Configuration FC is responsible for initializing the system's configuration.
 - 2) The Fault FC is used to identify, isolate, correct and log faults that occur in the IoT system.
 - 3) The Member FC is responsible for the management of the membership of any relevant entity
- The Reporting FC generates reports about the system and, finally, the State FC can change or enforce a particular state on the system by issuing a sequence of commands to the other FCs.

3.1.8 Operational View Specification

- Deployment and Operational View depends on the specific actual use case and requirements. Smart object in the IoT uses different methods for communication using different technology.
- Hence the Deployment and Operation view is very important to address how actual system can be realized by selecting technologies and making them communicate and operate in a comprehensive way.
- It provides an IoT Reference Model with a set of guidelines to application users. The different design choices that they have to face while designing the actual implementation of their services.
- The viewpoints used in the Deployment and Operation view are the following :
 - 1) The IoT Domain Model diagram is used as a guideline to describe the specific application domain.
 - 2) The Functional Model is used as a reference to the system definition.
 - 3) Network connectivity diagrams can be used to plan the connectivity topology to enable the desired networking capability of the target application; at the deployment level, the connectivity diagram will be used to define the hierarchies and the type of the sub-networks composing the complete system network;
 - 4) Device Descriptions can be used to map actual hardware on the service and resource requirements of the target system.

3.1.9 Device and Component Integration

- In this step we have to integrate the devices and components. The devices and components used in home automation examples are Raspberry Pi, sensor, laser pointer, light dependent resistor etc.
- Now is a good time to become acquainted with Raspberry Pi and its potential for use in IoT projects. When developing an innovative use for this device, you will need to come up with a software design solution that addresses your end-user requirements. You can accomplish this more easily when you have a better sense of the Raspberry Pi capabilities being demonstrated out in the field now.
- From controlling the room lights with your smartphone to scheduling events to occur automatically, home automation has taken convenience to a whole new level. Instead of using mechanical switches, you can now conveniently control all the devices in your home from your fingertips.

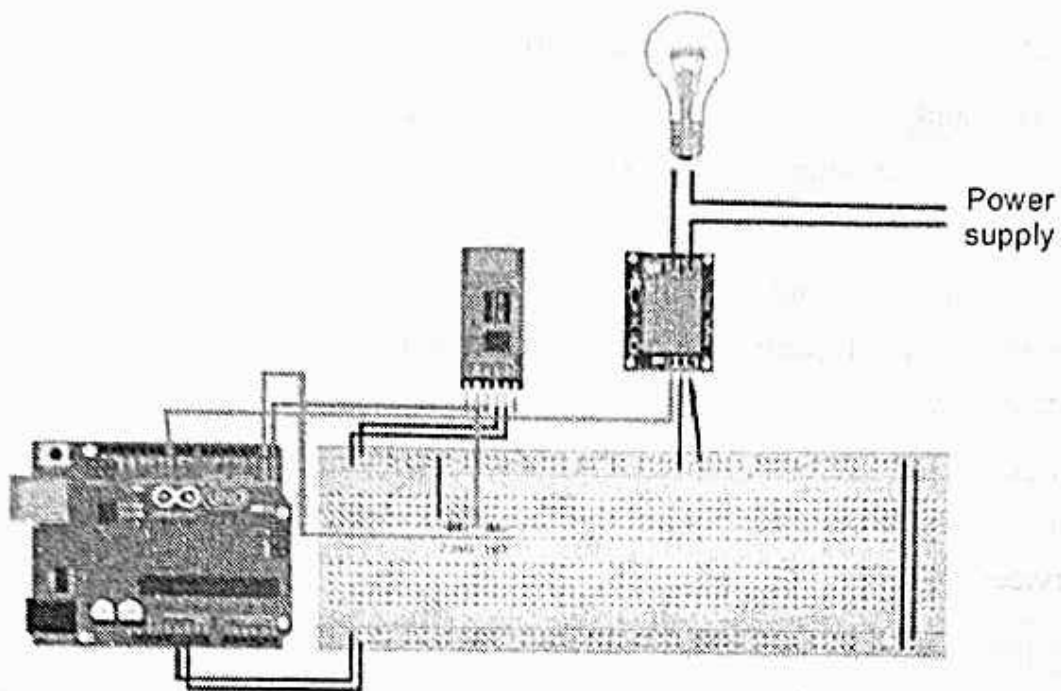


Fig. 3.1.9 : Schematic diagram for home automation

3.1.10 Application Development

- Application development is final stage in developing IoT application. In web application, it display mode control : Auto and Manual
- Two modes are provided : auto mode and manual mode.
- If auto mode is enabled the light control in the web application is disabled and it reflects the current state of light.
- If auto mode disabled, the light control is enabled and it is used for manually controlling light and air conditioner.

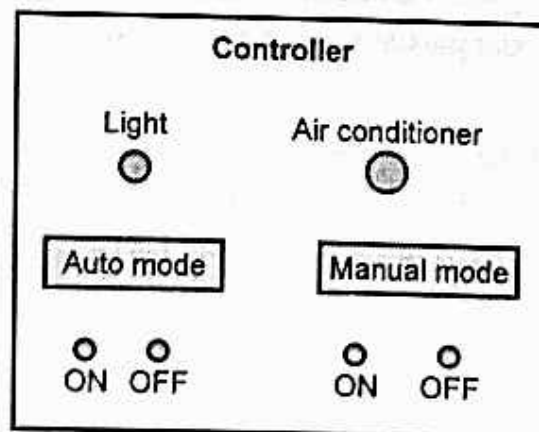


Fig. 3.1.10

Example 3.1.1 : Design smart irrigation system based on followings :

- i) Define process specification for smart irrigation IoT system
- ii) Domain model of smart irrigation IoT system
- iii) Information model of smart irrigation IoT system
- iv) Controller service of smart irrigation IoT system

SPPU - Dec.-19, End Sem, Marks 7

▣ **Solution :**

i) **Process specification :**

- Define the process with the help of use cases
- The use cases are formally described based on Purpose & requirement specification
- In this use case : Circle denotes a state or an attribute

ii) **Domain model**

- Describes the main concepts, entities and objects in the domain of IoT system to be designed
- It defines the attributes of the objects and relationships between them
- Entities , Objects and Concepts include the following : Physical entity, Virtual entity , Device, Resource, Service

Physical Entity :

- Discreet identifiable entity in physical environment
- For example : Pump, motor, LCD
- The IoT System provides the information about the physical entity (using sensors) or performs actuation upon the Physical entity (like switching a motor on etc.)
- In smart irrigation example, there are three Physical entities involved :
 - a. Soil (whose moisture content is to be monitored)
 - b. Motor (to be controlled)
 - c. Pump (To be controlled)
- In smart irrigation system there are three services :
 1. A service that sets the signal to low/ high depending upon the threshold value
 2. A service that sets the motor state on / off
 3. A controller service that runs and monitors the threshold value of the moisture and switches the state of motor on / off depending upon it.

- When threshold value is not crossed the controller retrieves the motor status from database and switches the motor on/off.

iii) Information Model Specification

- Defines the structure of all the information in the IoT system (such as attributes, relations etc.)
- It does not describe the specifics of how the information is represented or stored.
- This adds more information to the Virtual entities by defining their attributes and relations

iv) Controller service

- Define the services in IoT System, service types, service inputs / outputs, service endpoints, service schedules, service preconditions and service effects
- Services can be controller service, Threshold service, state service for smart irrigation system
- These services either change the state/attribute values or retrieve the current values.
- For eg :
 - a. Threshold service sets signal to high or low depending upon the soil moisture value.
 - b. State service sets the motor state : on or off
 - c. Controller service monitors the threshold value as well as the motor state and switches the motor on / off and updates the status in the database.

Review Questions

1. Explain the steps involved in the IoT system design methodology.

SPPU : May-18, Marks 4

2. Explain functional view specification step of IoT system design methodology, consider smart IoT-based home automation system as an example.

SPPU : May-18, Dec.-19, Marks 7

3. Explain service specification step of IoT system, design methodology, consider smart IoT - based home automation system as an example.

SPPU : Dec.-18, March-20, Marks 5

4. Explain operational view specification step of IoT system design methodology, consider smart IoT - based home automation system as an example.

SPPU : Dec.-18, Marks 6

5. Explain purpose and requirement specification step of IoT system design methodology, consider smart IoT-based automation system as an example.

SPPU : May-19, Marks 5

6. What is the importance of service specification in IoT design methodology ?

SPPU : May-19, Marks 4

7. Explain domain model specification step of IoT system design methodology, consider smart IoT-based home automation system as an example.

SPPU : Dec.-19, Marks 4

8. With the help of diagram list and briefly explain the steps involved in the IoT system design methodology.

SPPU : March-20, Marks 5

9. Explain process specification for home automation IOT system.

SPPU : March-20, Marks 5

10. Explain the operational view specification step of IoT system design.

SPPU : March-20, Marks 5

3.2 Basic of IoT Networking

- An IoT network refers to a collection of interconnected devices that communicate with other devices without the need for human involvement, such as autonomous cars, smart appliances, and wearable technology. Fig. 3.2.1 shows IoT network.

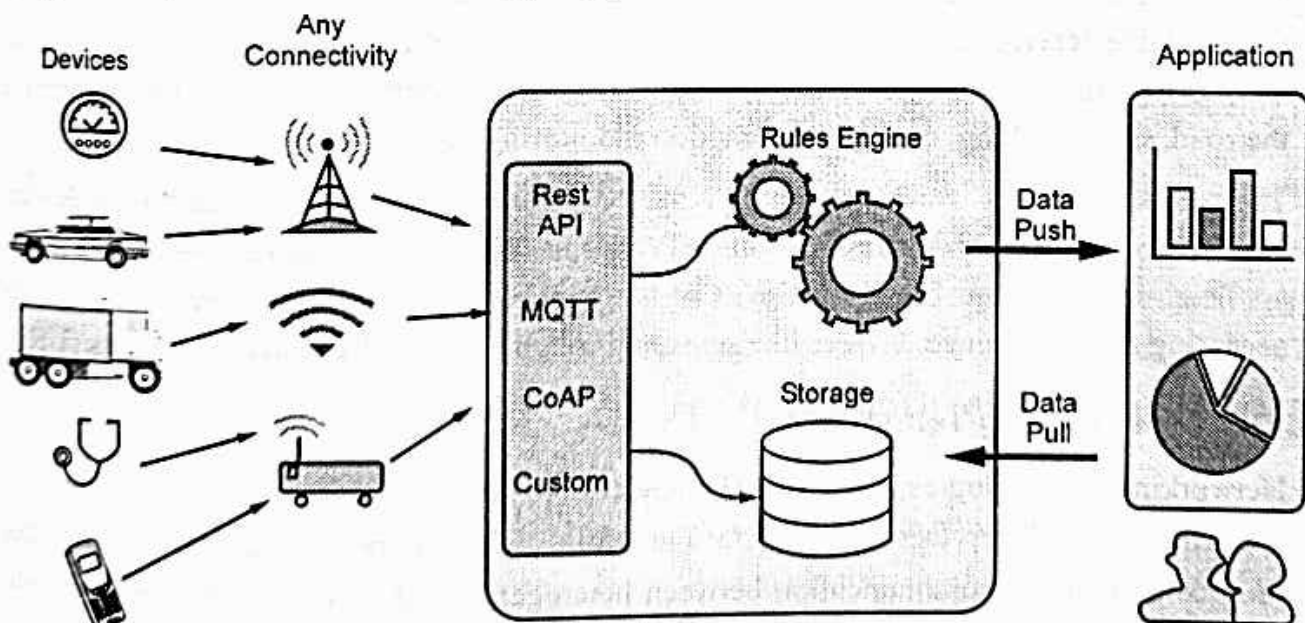


Fig. 3.2.1 : IoT Network

- The main aim of IoT is to make Internet more and more inspiring and pervasive. By connecting wide variety of heterogeneous devices and making easy access of the devices, several applications are being used in IoT, which deals with the large number of data generated by the attached devices and make some decisions which is very important for the industry or control the attached devices based on the generated data.
- In the Internet of Things, all the things can be put into three categories :
 1. Sensors that collect information and then send it.
 2. Computers that receive information and then act on it.
 3. Things that do both.
- Sensors, when paired with an internet connection, allow us to collect information from the environment which, in turn, helps make better decisions.
- With the advancement of communication technology and microcontroller technology, Wireless Sensor Network (WSN) plays an important role in IoT.
- Sensor nodes, which are very tiny low cost devices, can sense different parameters in the environment, it can store data in its memory temporarily and can communicate with other electronics devices by using communication technology like IEEE802.15.4 or Bluetooth Low Energy (BLE).
- On the other hand RFID tag is also a useful device in IoT, which is used to identify an object attached with a RFID tag.
- Smart city is one of the most demanding and growing applications of IoT. In a smart city, the available services are mainly structural health of the building or bridge, noise pollution monitoring, air pollution monitoring, waste management monitoring, traffic management in the road, smart parking, energy consumption monitoring etc.
- The web applications which are mainly XML based are not suitable for constrained devices like sensor or RFID devices. So in IoT, people have started developing web based applications which are based on CoAP in the application layer. It has been provided for accessing the constrained devices like sensors through Uniform Resource Identifiers (URI).

3.2.1 Networking Tehnology

- Networking technologies enable IoT devices to communicate with other devices, applications, and services running in the cloud. The internet relies on standardized protocols to ensure communication between heterogeneous devices is secure and reliable. Standard protocols specify rules and formats that devices use to establish and manage networks and transmit data across those networks.

- Fig. 3.2.2 shows an IoT network protocols mapped to the TCP/IP model.

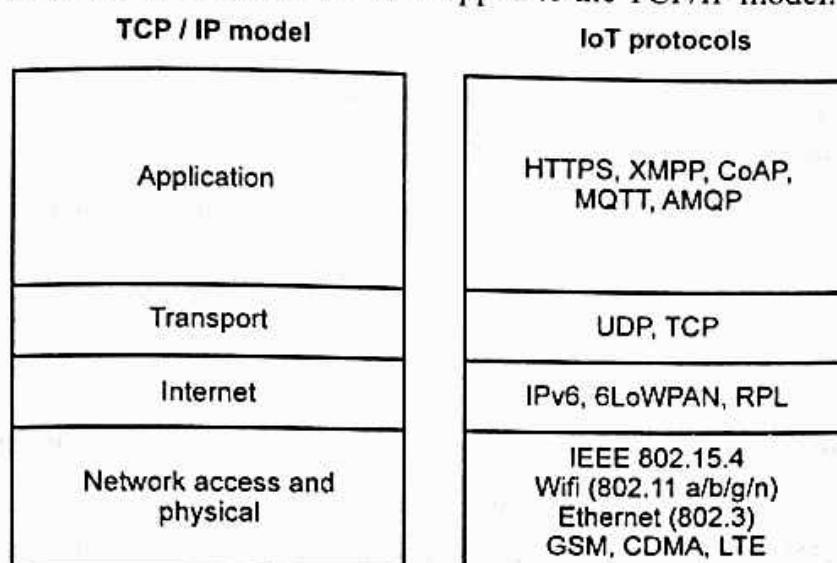


Fig. 3.2.2

- Technologies used for internet connectivity, such as Ethernet, can be applied within the IoT. The structure of a network is known as its topology. The most common network topologies that are adopted within IoT are star and mesh topologies.
 - IoT network technologies to be aware of toward the bottom of the protocol stack include cellular, wifi, and Ethernet, as well as more specialized solutions such as LPWAN, Bluetooth Low Energy (BLE), ZigBee, NFC, and RFID.
- 1. LPWAN (Low Power Wide Area Network) :** LPWAN is a category of technologies designed for low-power, long-range wireless communication. They are ideal for large-scale deployments of low-power IoT devices such as wireless sensors.
 - 2. Cellular :** The LPWAN NB-IoT and LTE-M standards address low-power, low-cost IoT communication options using existing cellular networks. NB-IoT is the newest of these standards and is focused on long-range communication between large numbers of primarily indoor devices. LTE-M and NB-IoT were developed specifically for IoT, however existing cellular technologies are also frequently adopted for long-range wireless communication.
 - 3. Bluetooth Low Energy :** BLE is a low-power version of the popular Bluetooth 2.4 GHz wireless communication protocol. It is designed for short-range communication, typically in a star configuration, with a single primary device that controls several secondary devices.

4. **ZigBee** : ZigBee operates on 2.4GHz wireless communication spectrum. It has a longer range than BLE by up to 100 meters. It also has a slightly lower data rate than BLE. ZigBee is a mesh network protocol. ZigBee was designed for building and home automation applications.
5. **NFC** : The Near Field Communication (NFC) protocol is used for very small range communication (up to 4 cm), such as holding an NFC card or tag next to a reader. NFC is often used for payment systems, but also useful for check-in systems and smart labels in asset tracking.
6. **RFID (Radio Frequency Identification)** : RFID tags store identifiers and data. The tags are attached to devices and read by an RFID reader. The typical range of RFID is less than a meter. RFID tags can be active, passive, or assisted passive. Passive tags are ideal for devices without batteries, as the ID is passively read by the reader. Active tags periodically broadcast their ID, while assisted passive tags become active when RFID reader is present.
7. **Wifi** : Wifi is standard wireless networking based on IEEE 802.11a/b/g/n specifications. 802.11n offers the highest data throughput, but at the cost of high-power consumption, so IoT devices might only use 802.11b or g for power conservation reasons.
8. **Ethernet** : Widely deployed for wired connectivity within local area networks, Ethernet implements the IEEE 802.3 standard. Not all IoT devices need to be stationery wireless . For example, sensor units installed within a building automation system can use wired networking technologies like Ethernet. Power line communication (PLC), an alternative hard-wired solution, uses existing electrical wiring instead of dedicated network cables.

3.3 Internet Structure

- The internet is a global network of interconnected networks that communicate using a common set of standards and protocols. These networks are owned and managed by a wide range of organisations, such as national governments, private companies, and academic institutions.
- Individuals and organisations connect to the internet through an Internet Service Provider (ISP). Every host is assigned, and identified by, an IP address. Each packet contains a header that specifies the destination address. The network routes the packets from the source to the destination.

- Fig. 3.3.1 shows high-level structure of the Internet.

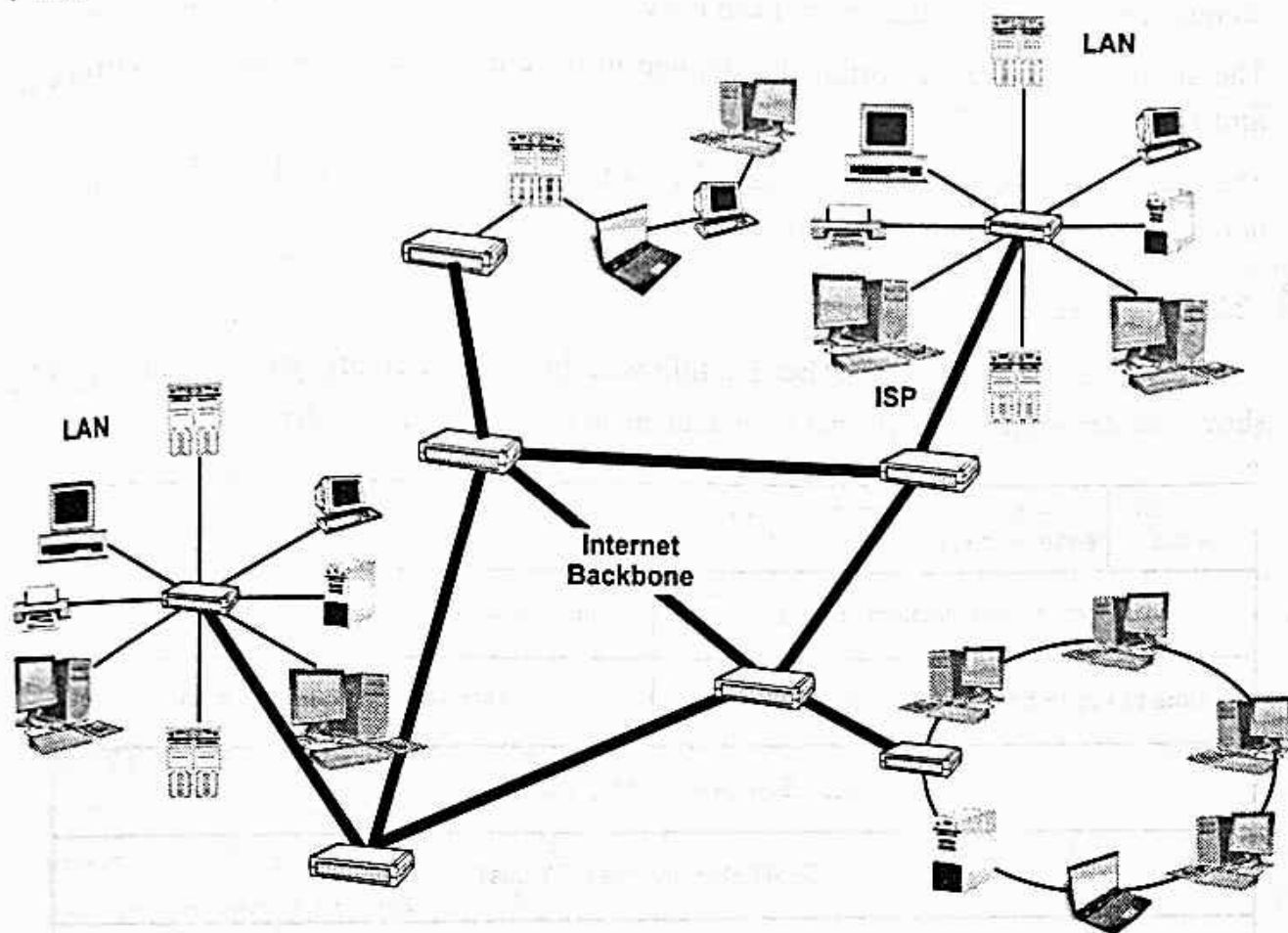


Fig. 3.3.1

- The internet consists of a hierarchy of networks, interconnected via a network backbone. Each network has a unique network address.
- Computers, or hosts, are connected to a network. Each host has a unique ID within its network. Each process running on a host is associated with zero or more ports. A port is a logical entity for data transmission.

3.3.1 IPv4

- IPv4 is Internet Protocol version 4. It is the network layer protocol of the TCP/IP protocol suite.
- IP is a connectionless, unreliable, best-effort delivery protocol. All the nodes are identified using an IP address. Packets are delivered from the source to the destination using IP address.
- IPv4 addresses are encoded as a 32 bits field. IPv4 addresses are often represented in dotted-decimal format as a sequence of four integers separated by a dot.

- An IPv4 address is used to identify an interface on a router or a host. IPv4 addresses are unique. Two devices on the internet can never have the same address at the same time.
- The address structure was originally defined to have a two-level hierarchy : **Network ID and host ID**.
- The network ID identifies the network the host is connected to. The host ID identifies the network connection to the host rather than the actual host.

3.3.1.1 Packet Format

- The IP datagram consists of a header followed by a number of bytes of data. Fig 3.3.2 shows header format of IPv4. All IPv4 packets use the 20 bytes header.

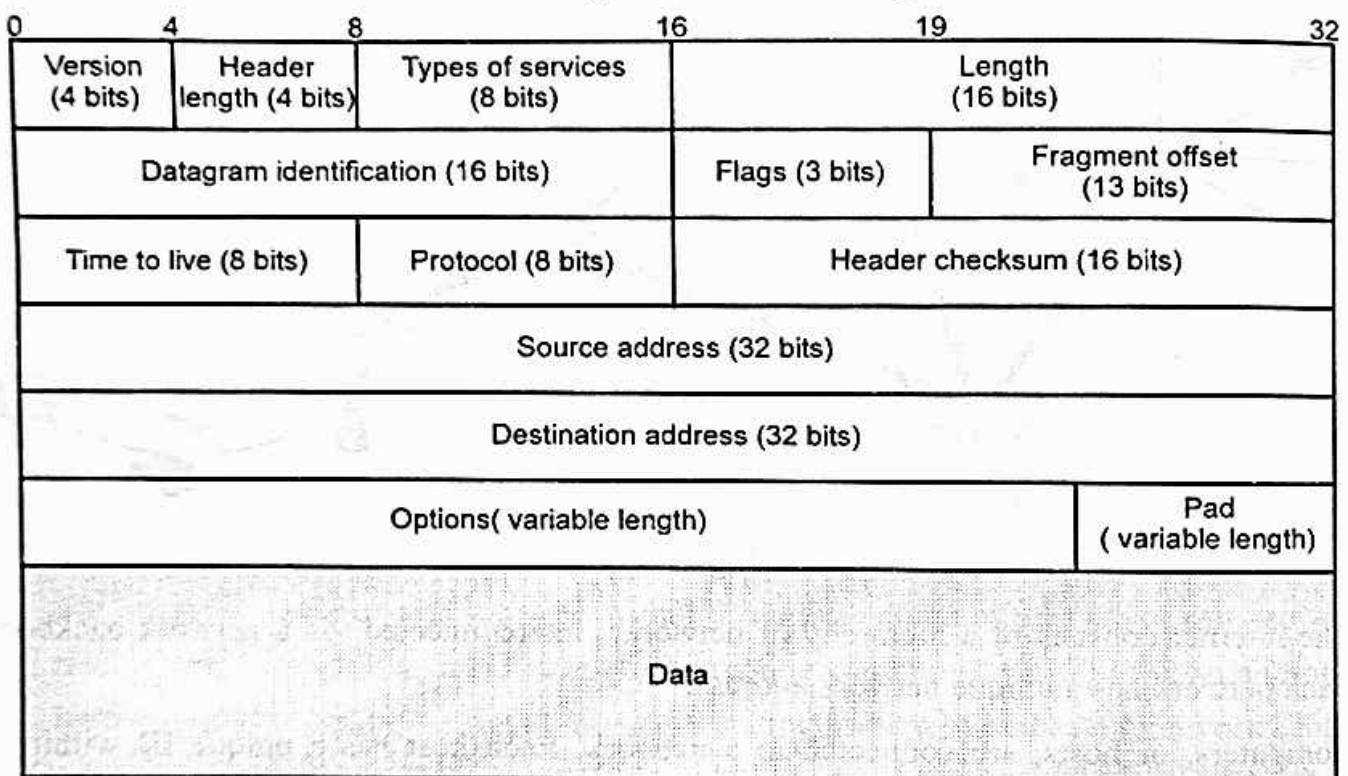


Fig. 3.3.2 : IPv4 header format

- **Version field** indicates the version of IP used to build the header. Current version is 4.
- **Header Length** indicates the length of the IP header in 32 bits words. This field allows IPv4 to use options if required, but as it is encoded as a 4 bits field, the IPv4 header cannot be longer than 64 bytes.
- **Type of service** : The service type is an indication of the quality of service requested for this IP datagram.
- **Protocol field** indicates the transport layer protocol that must process the packet's payload at the destination. Common values for this field are 6 for TCP and 17 for UDP

- **Length field** indicates the total length of the entire IPv4 packet (header and payload) in bytes. This implies that an IPv4 packet cannot be longer than 65535 bytes.
- **Fragment offset** is used to reassemble the full datagram. The value in this field contains the number of 64-bit segments (header bytes are not counted) contained in earlier fragments. If this is the first (or only) fragment, this field contains a value of zero.
- **Flags (3 bits)** : Only two of the bits are currently defined : MF (More Fragments) and DF (Don't Fragment).
- **More Fragments (MF) flag** is a single bit in the Flag field used with the Fragment Offset for the fragmentation and reconstruction of packets. The More Fragments flag bit is set, it means that it is not the last fragment of a packet. When a receiving host sees a packet arrive with the MF = 1, it examines the Fragment Offset to see where this fragment is to be placed in the reconstructed packet. When a receiving host receives a frame with the MF = 0 and a non-zero value in the Fragment offset, it places that fragment as the last part of the reconstructed packet.
- **Don't Fragment (DF) flag** is a single bit in the Flag field that indicates that fragmentation of the packet is not allowed. If the Don't Fragment flag bit is set, then fragmentation of this packet is NOT permitted. If a router needs to fragment a packet to allow it to be passed downward to the Data Link layer but the DF bit is set to 1, then the router will discard this packet.
- **Time-to-Live (TTL)** is an 8-bit binary value that indicates the remaining "life" of the packet. The TTL value is decreased by at least one each time the packet is processed by a router (that is, each hop). When the value becomes zero, the router discards or drops the packet and it is removed from the network data flow.
- **Source address field** that contains the IPv4 address of the source host.
- **Destination address field** that contains the IPv4 address of the destination host
- **Checksum** that protects only the IPv4 header against transmission errors. Checksum is for the information contained in the header. If the header checksum does not match the contents, the datagram is discarded.
- **Options (variable)** : Encodes the options requested by the sending user.
- **Padding (variable)** : Used to ensure that the datagram header is a multiple of 32 bits.
- **Data (variable)** : The data field must be an integer multiple of 8 bits. The maximum length of the datagram (data field plus header) is 65,535 octets.

3.3.1.2 Classes of IP Address

- An IPv4 address is a 32-bit sequence of ones and zeros. To make the IP address easier to work with, it is usually written as four decimal numbers separated by periods.
- For example, an IP address of one computer is 192.168.1.2. This is called the dotted decimal format.
- Each part of the address is called an octet because it is made up of eight binary digits. For example, the IP address 192.168.1.8 would be 11000000.10101000.00000001.00001000 in binary notation.
- Address 0.0.0.0, 127.0.0.1 and 255.255.255.255 carries special meaning. IP address is divided into a network number and a host number. The network prefix identifies a network and the host number identifies a specific host.
- Example :

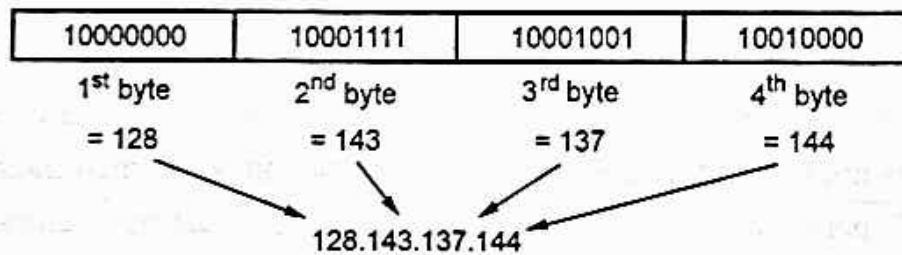


Fig. 3.3.3

- The IP address consists of a pair of numbers : IP address = <network number><host number>
- Example : Suppose IP address of technicalpublication.org is 129.144.136.146, then network address is 129.144.0.0 and host number is 136.146.

Classes of IP Address

- IP address is divided into five classes : A, B, C, D, and E. Fig. 3.3.4 shows IP address classes with net ID and host ID.
- The class of an IP address is identified in the most significant few bits. If the first bit is 0, it is a class A address.
- If the first bit is 1 and second is 0, it is a class B address. If the first two bits are 1 and third is 0, it is a class C address.
- Class D is used for what is known as multicasting, transmitting data to multiple computers or routers.
- Class E was reserved for future use, but this has given way to IPv6 instead.

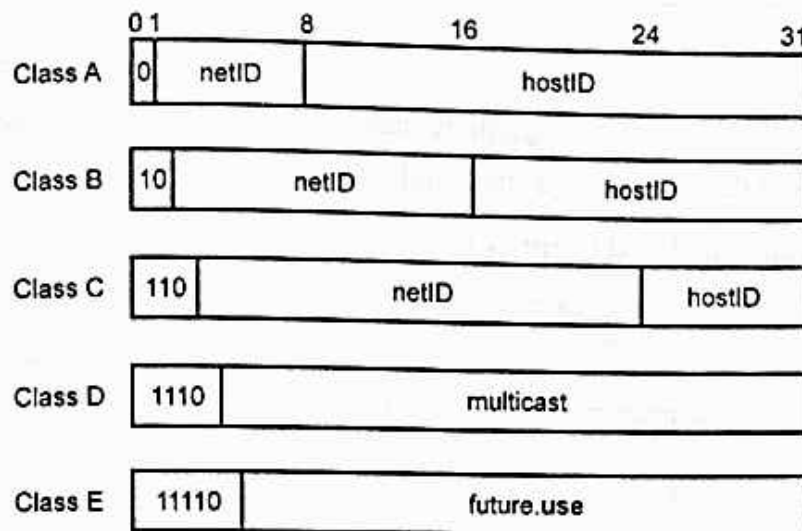


Fig. 3.3.4 : IP address classes with net ID and host ID

• Range of IP address :

Class	Starting address	Last Address
Class A	0.0.0.0	127.255.255.255
Class B	128.0.0.0	191.255.255.255
Class C	192.0.0.0	223.255.255.255
Class D	224.0.0.0	239.255.255.255
Class E	240.0.0.0	255.255.255.255

- **Loopback Testing :** The 127 network number isn't used by hosts as a **logical IP address**. Instead, this network is used for **loopback IP addresses**, which allow for testing.
- Address 0.0.0.0, 127.0.0.1 and 255.255.255.255 carries special meaning.



3.3.1.3 Public and Private Addresses

- IPv4 addresses are further classified as either public or private. Public IP addresses are ones that are exposed to the Internet. Any other computers on the Internet can potentially communicate with them.
- Private IP addresses are hidden from the Internet and any other networks. They are usually behind an IP proxy or firewall device.
- Private Addresses are as follows :

Class	Starting Address	End of range
Class A	10.0.0.0	10.255.255.255
Class B	172.16.0.0	172.31.255.255
Class C	192.168.0.0	192.168.255.255

□ Subnetting

- Subnetting is the subdivision of logical IP network. By default, all computers are on one subnet or network with no divisions involved.
- Subnet mask of class A, B and C are as follows :

IP address Class	Subnet Mask
A	255.0.0.0
B	255.255.0.0
C	255.255.255.0



3.3.2 IPv6

- IPv4 provides the host to host communication between systems in the Internet.
- IPv4 has played a central role in the internetworking environment for many years. It has proved flexible enough to work on many different networking technologies.
- In the early 1990 the IETF began to work on the successor of IPv4 that would solve the address exhaustion problem and other scalability problems.
- IPv6 addresses are 128 bits in length. Addresses are assigned to individual interface on nodes, not to the node themselves.
- IPv6 addresses are assigned to interfaces, rather than to nodes, in recognition that a node can have more than one interface.
- A single interface may have multiple unique unicast addresses. The first field of any IPv6 address is the variable length format prefix, which identifies various categories of addresses.
- A new notation has been devised for writing 16-byte addresses. They are written as eight groups of four hexadecimal digits with colons between the groups, like this

8000 : 0000 : 0000 : 0000 : 0123 : 4567 : 89AB : CDEF

□ Optimization

1. Leading zeros within a group can be omitted so 0123 can be written as 123.
2. One or more groups of 16 zero bits can be replaced by a pair of colons. The address new becomes

8000 :: 123 : 4567 : 89AB : CDEF

□ Address Types

- IPv6 allows three types of addresses : Unicast, Anycast and Multicast.
- **Unicast** : An identifier for a single interface. A packet sent to a unicast address is delivered to the interface identified by that address.
- **Anycast** : An identifier for a set of interfaces. A packet sent to an anycast address is delivered to one of the interfaces identified by the address.
- **Multicast** : An identifier for a set of interfaces. A packet sent to a multicast address is delivered to all interfaces identified by that address.
- The first field of any IPv6 address is the variable-length format prefix, which identifies various categories of address.

📄 3.3.2.1 Packet Format

- Fig. 3.3.5 the IPv6 datagram header format. Each packet is composed of a mandatory base header followed by the payload.

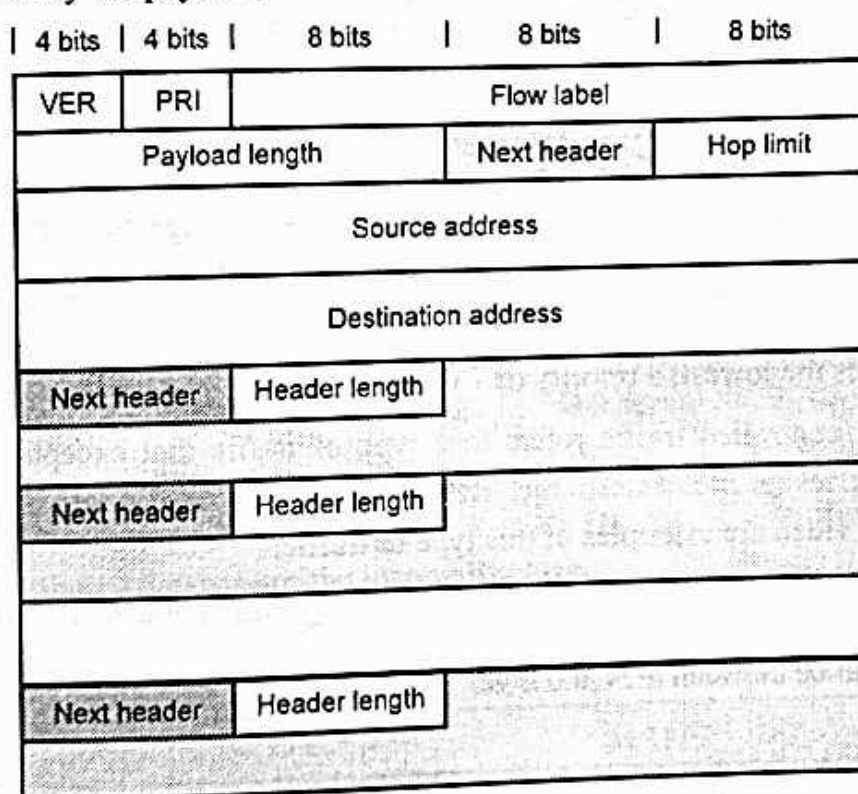


Fig. 3.3.5 : IPv6 header

- The payload consists of two parts : Optional and data.
 1. Versions : This 4 bits field defines the version number of the IP. The value is 6 for IPv6.
 2. Priority : The 4 bits priority field defines the priority of the packet with respect to traffic congestion.

3. Flow label : It is 24 bits field that is designed to provide special handling for a particular flow of data.
4. Payload length : The 16 bits payload length field defines the length of the IP datagram excluding the base header.
5. Next header : It is an 8 bits field defining the header that follows the base header in the datagram.
6. Hop limit : This 8 bits hop limit field serves the same purpose as the TTL field in IPv4.
7. Source address : The source address field is a 128 bits internet address that identifies the original.
8. Destination address : It is 128 bits Internet address that usually identifies the final destination of the datagram.

Priority :

- The priority field defines the priority of each packet with respect to other packets from the same source. IPv6 divides traffic into two broad categories
 1. Congestion controlled
 2. Noncongestion controlled
- If a source adapts itself to traffic slowdown when there is congestion, the traffic is referred to as congestion controlled traffic congestion controlled data are assigned priorities from 0 to 7.
- A priority of 0 is the lowest; a priority of 7 is the highest.
- Noncongestion controlled traffic refers to a type of traffic that expects minimum delay. Discarding of packets is not desirable. Retransmission in most cases is impossible. Real time audio and video are examples of this type of traffic.
- Priority numbers from 8 to 15 are assigned to noncongestion controlled traffic.

3.3.2.2 Difference between IPv4 and IPv6

Sr. No.	IPv4	IPv6
1.	Header size is 32 bits.	Header size is 128 bits.
2.	It cannot support autoconfiguration.	Supports autoconfiguration.
3.	Cannot support real time application.	Supports real time application.
4.	No security at network layer.	Provides security at network layer.
5.	Throughput and delay is more.	Throughput and delay is less.

3.3.2.3 Advantages of IPv6

1. Larger address space
2. Better header format
3. Security capabilities
4. Support for resource allocation
5. New options
6. Allowance for extension

3.3.3 Domain Name System (DNS)

- The DNS is a distributed database that resides on multiple machines on the internet and used to convert between names and address and to provide e-mail routing information.
- For user friendliness, each Internet address is mapped to a symbolic name, using the DNS, in the format of :

<computer name>.<subdomain hierarchy>.<organization>.<sector name>{.<country code>}

e.g., www.technicalpublications.org

- Fig. 3.3.6 shows the DNS in the internet.

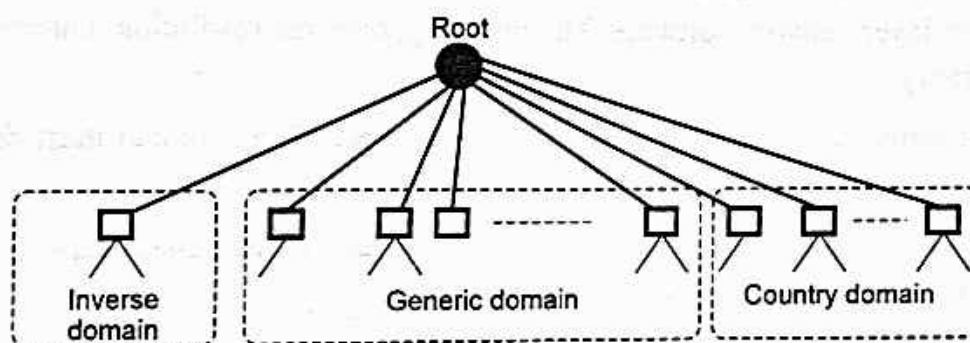


Fig. 3.3.6 : The DNS In the Internet

- For network applications, a domain name must be mapped to its corresponding Internet address. Processes known as domain name system servers provide the mapping service, based on a distributed database of the mapping scheme.
- The mapping service is offered by thousands of DNS servers on the Internet, each responsible for a portion of the name space, called a zone. The servers that have access to the DNS information (zone file) for a zone is said to have authority for that zone.
- The Domain Name System (DNS) is a hierarchical, distributed naming system designed to cope with the problem of explosive growth :
 1. It is hierarchical because the name space is partitioned into subdomains.
 2. It is distributed because management of the name space is delegated to local sites. Local sites have complete control (and responsibility) for their part of the name space. DNS queries are handled by servers called name servers.

3. It does more than just map machine names to internet addresses. For example, it allows a site to associate multiple machines with a single, mailbox name.

- In the DNS, the name space is structured as a tree, with domain names referring to nodes in the tree. The tree has a root and a fully-qualified domain name is identified by the components of the path from the domain name to the root.
- In zone, a server is responsible and have some authority. The server makes database called zone file and keeps all the information for every node under that domain.
- Domain and zone are same if server accepts responsibility for a domain and does not divide the domain into subdomain.
- Domain and zone are different, if a server divides its domain into subdomains and delegates part of its authority to other server.

☐ Name lookup and resolution

- If a domain name is used to address a host, its corresponding IP address must be obtained for the lower-layer network software. The mapping, or name resolution, must be maintained in some registry.
- For runtime name resolution, a network service is needed; a protocol must be defined for the naming scheme and for the service.
- **Example :** The DNS service supports the DNS; the Java RMI registry supports RMI object lookup; JNDI is a network service lookup protocol.
- DNS is designed as a client server application. A host that needs to map an address to a name or a name to an address calls a DNS client named a resolver.
- Name resolving must also include the type of answer desired. The DNS partitions the entire set of names by class (for mapping to multiple protocol suites).
- Naming items is required since one cannot distinguish the names of subdomains from the names of individual objects or their types.

☐ Well Known Ports :

- Each Internet host has 2^{16} (65,535) logical ports. Each port is identified by a number between 1 and 65535 and can be allocated to a particular process.
- Port numbers between 1 and 1023 are reserved for processes which provide well-known services such as finger, FTP, HTTP, and email.
- Port numbers from 0 to 65535 is used in Internet. It is 16 bits integer so the range is 0 to 65535. The client program defines itself with a port number, chosen randomly by the transport player software running on the client host. This is the ephemeral port number.

- Server also define a port number but not randomly. Internet has decided to use universal port numbers for servers, these are called well known port numbers. The port number ranging from 0 to 1023 are called well known port numbers and are restricted, which means that they are reserved for use by well known application protocols such as HTTP.
- The Internet Assigned Number Authority (IANA) has divided the port number into three ranges. They are :

a) Well known ports

b) Registered ports

c) Dynamic ports

Sr. No.	Port type	Range	Remark
1.	Well known port	0 to 1023	Assigned and controlled by IANA.
2.	Registered port	1024 to 49151	Not assigned and controlled by IANA. Only registered to prevent duplication.
3.	Dynamic	49152 to 65535	Neither controlled nor registered. Used by any process. These are ephemeral ports.

3.3.4 The Uniform Resource Identifier (URI)

- Resources to be shared on a network need to be uniquely identifiable. On the Internet, a URI is a character string which allows a resource to be located.
- There are two types of URIs :
 - a) URL (Uniform Resource Locator) points to a specific resource at a specific location
 - b) URN (Uniform Resource Name) points to a specific resource at a nonspecific location.

Uniform Resource Locator

- The Uniform Resource Locator (URL) is a standard for specifying any kind of information on the Internet.
- URL has three parts :
 1. The protocol
 2. DNS name of the machine where the page is located
 3. File name containing the page.
- URL is represented as

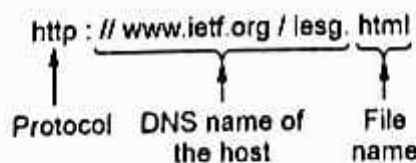


Fig. 3.3.7

- The protocol is the client-server program used to retrieve the document.
- Host is the computer on which the information is located.
- The URL can optionally contain the port number of the server.
- File name gives where the information is located.

3.4 IoT Communication Model

➡ SPPU : May-19

Request/Response model :

- In the Request/Response model, client requests information from the server and waits till the response is served from the server. Fig. 3.4.1 shows Request/Response model.

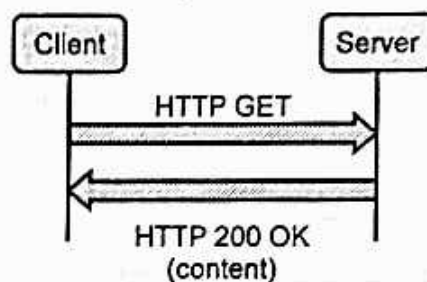


Fig. 3.4.1 : Request/Response model

- HTTP protocol is used by Request / Response model. For example, a browser client may request a web page from the server through a "Request" and the corresponding web page will be served by the server as a "Response".
- The client and the server can communicate one to one, or one to many with more requests.
- This model is stateless communication model and each request-response pair is independent of others.

Publish/Subscribe Model :

- Publishers : Publishers generate event data and publishes them.
- Subscribers : Subscribers submit their subscriptions and process the events received
- Publish / Subscribe service : It's the mediator / broker that filters and routes events from publishers to interested subscribers.
- Fig. 3.4.2 shows Publish / Subscribe Model.
- The publishers and subscribers are autonomous, which means that they do not need to know the presence of each other.
- This model is highly suited for mobile applications, ubiquitous computing and distributed embedded systems.

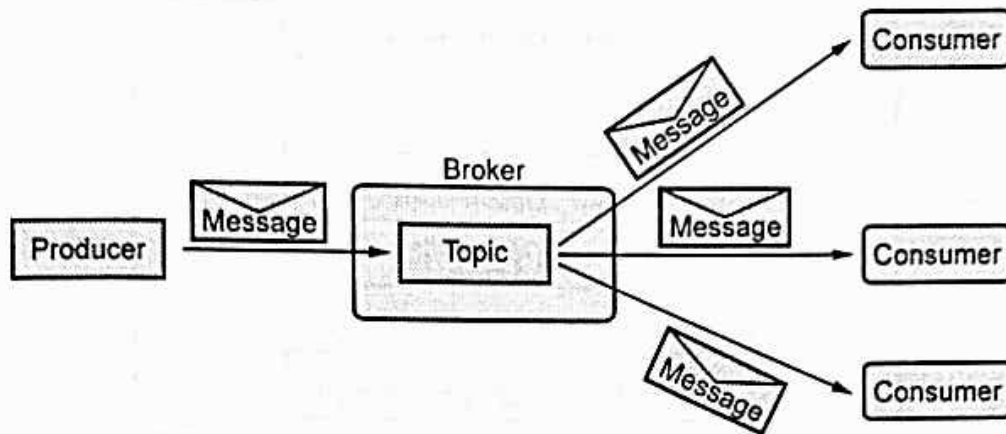


Fig. 3.4.2 : Publish/Subscribe model

- Failure of publishers or subscribers does not bring down the entire system.
- No strong guarantee on broker to deliver content to subscriber. After a publisher publishes the event, it assumes that all corresponding subscribers would receive it.
- Potential bottleneck in brokers when subscribers and publishers overload them.

❑ Push / Pull Model :

- Data procedure push the data to queues and consumers pull the data from the queues. Fig. 3.4.3 shows push-pull model.

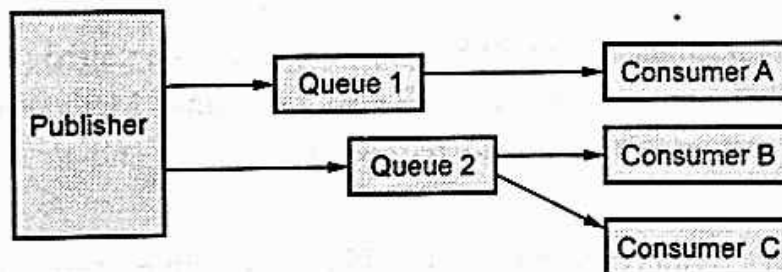


Fig. 3.4.3 : Push-Pull model

- Sometimes queue act as buffer in between producer and consumer. Producer does not need to be aware of the consumers.

❑ Exclusive Pair Model :

- This communication model is full duplex, bi-directional communication model. It uses persistent connection between client and server. Fig. 3.4.4 shows exclusive pair model.
- Client send request to server for opening the connection. This connection is open till the client send request for closing the connection.

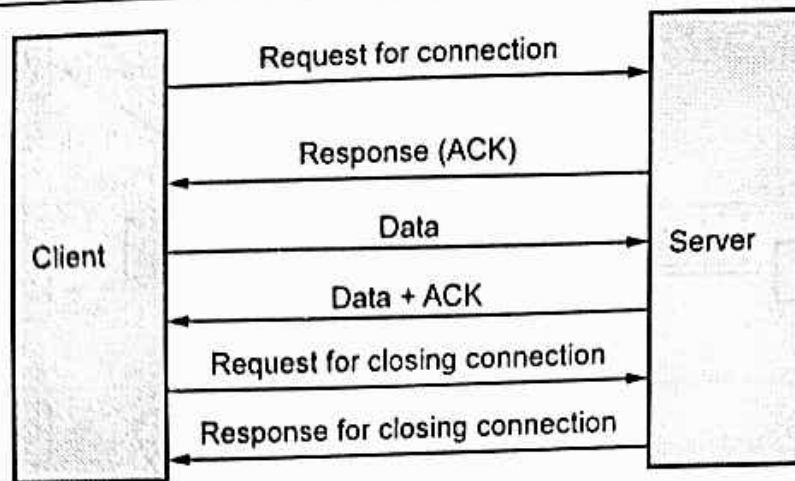


Fig. 3.4.4 : Exclusive pair model

3.4.1 IoT Communication APIs

1. REST Based Communication API :

- A large part of the interoperability, scale, and control for IoT can be achieved through API management. Standards-based design patterns for Web APIs, API management, and a RESTful architecture provide tremendous value in simplifying the task of interoperability across heterogeneous systems handling vast amounts of data.
 - Representational State Transfer (REST) APIs follow the request-response communication model.
 - Applications conforming to the REST constraints can be called RESTful. RESTful systems typically communicate over HTTP with the same Methods (GET, POST, PUT, DELETE etc) that browsers use to retrieve web pages and to send data to remote servers.
1. Client-Server : requires that a service offer one or more operations and that services wait for clients to request these operations.
 2. Stateless : requires communication between service consumer (client) and service provider (server) to be stateless.
 3. Cache : requires responses to be clearly labeled as cacheable or non-cacheable.
 4. Uniform Interface : requires all service providers and consumers within a REST-compliant architecture to share a single common interface for all operations.
 5. Layered System : requires the ability to add or remove intermediaries at runtime without disrupting the system.
 6. Code-on-Demand : allows logic within clients (such as Web browsers) to be updated independently from server-side logic using executable code shipped from service providers to consumers.

- Each client request and server response is a message, and REST-compliant applications expect each message to be self-descriptive. That means each message must contain all the information necessary to complete the task. Other ways to describe this kind of message are "stateless" or "context-free." Each message passed between client and server can have a body and metadata.
- **HTTP request methods and actions :**
 1. GET : return whatever information is identified by the Request-URI
 2. HEAD : identical to GET except that the server must not return a message-body in the response, only the metadata
 3. OPTIONS : return information about the communication options available on the request/response chain identified by the Request-URI
 4. PUT : requests that the enclosed entity be stored under the supplied Request-URI
 5. POST : requests that the origin server accept the entity enclosed in the request as a new subordinate of the resource identified by the Request-URI
 6. DELETE : requests that the origin server delete the resource identified by the Request-URI.
- The first three are read-only operations, while the last three are write operations.

❑ 2. WebSocket based Communication API

- WebSocket support full-duplex, two-way communication between client and server.
- WebSocket APIs reduce the network traffic and latency as there is no overhead for connection setup and termination requests for each message.
- WebSocket uses a standard HTTP request-response sequence to establish a connection. When the connection is established, the WebSocket API provides a read and write interface for reading and writing data over the established connection in an asynchronous full duplex manner.
- WebSocket also provides an interface for asynchronously closing the connection from either side.

Review Question

1. What is publisher-subscriber communication model with diagram ?

SPPU : May-19, Marks 5

3.5 Sensor Network

- A Wireless Sensor Network (WSN) is composed of autonomous devices called sensor nodes that generally have low computational power, limited data transmission and power constraints.
- A WSN consists of sensor nodes that capturing information from an environment, processing data and transmitting them via radio signals.
- Generally, WSNs have no fixed structure, and in many cases there is no monitoring station of sensor nodes during the operational life of the network, so a WSN must have mechanisms for self-configuration and adaptation in case of failure, inclusion or exclusion of a sensor node.
- Fig. 3.5.1 shows sensor network.

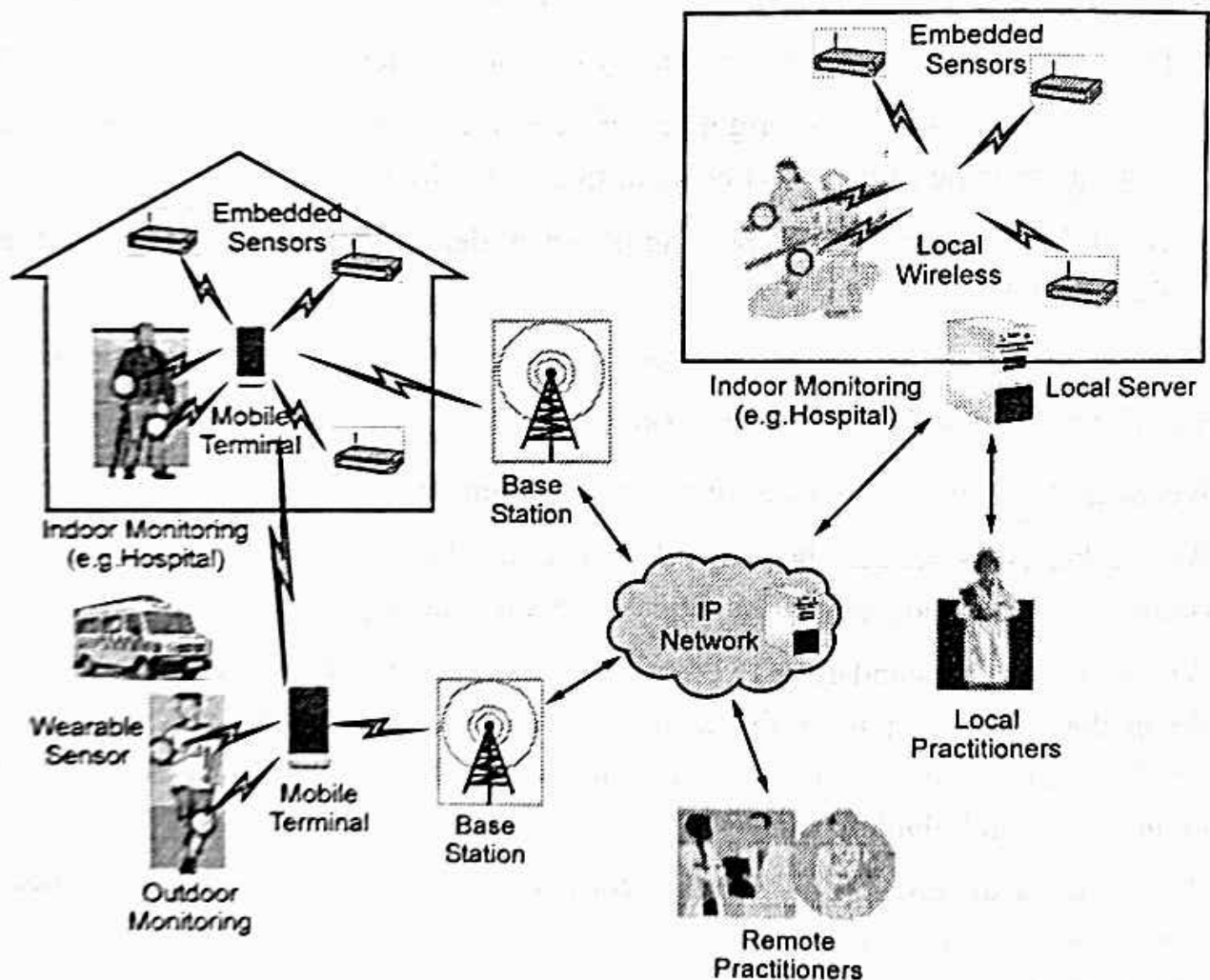


Fig. 3.5.1 : Sensor network

- Sensor devices are basically formed by a computational part responsible for storing and transmitting data, and a sensing portion which can be formed by one or more sensors, such as acoustic, seismic, infrared video camera, heat temperature and pressure.

- In general, two modulation formats are available: Frequency-Shift-Keyed (FSK) operating at 433 and 868-915 MHz and Direct Sequence Spread Spectrum (DSSS) operating at 2.4 GHz band that transmit 802.15.4 and ZigBee standards.
- The reach of the radios varies from 10 to 100 meters. The antenna configuration can cause transmission rates from 19.2 kbps to 240 kbps.
- Currently, the sensor nodes can vary between mode activity, inactivity (idle) and low consumption (sleep) in order to save energy. The energy issue is important because most of sensor nodes are powered by batteries.
- Sensors usually communicate with each other using a multi hop approach.
- A base station links the sensor network to another network (like a gateway) to disseminate the data sensed for further processing. Base stations have enhanced capabilities over simple sensor nodes since they must do complex data processing.
- Aggregation points are regular nodes that receive data from neighboring nodes, perform some kind of processing, and then forward the filtered data to the next hop.
- Sensor nodes are organized into clusters, each cluster having a "cluster head" as the leader. The communication within a cluster must travel through the cluster head, which is then forwarded to a neighboring cluster head until it reaches its destination, the base station.
- WSNs can be used in following situations :
 1. **Environment** - Monitoring of environmental variables such as buildings, residences and external locations such as oceans, volcanoes, deserts, etc.
 2. **Traffic** - Monitoring of vehicle traffic on highways, railroads, rivers, oceans, etc.
 3. **Security** - To provide security in homes, shopping centers, farms, among others.
 4. **Military** - To detect the presence of enemies, explosions, presence of hazardous materials as poison gas and radiation.

☐ **Security vulnerabilities :**

- In most of applications, sensor devices are spread over large areas, what difficult an individual control of network components. Moreover, wireless communication allows an attacker can trigger attacks without having physical access to the device.
- Attacks on WSNs can be divided into three main types :
 - (1) **Attack of authentication and confidentiality:** Consists of attacks change, repetition or modification packages.

- (2) Availability network Attack : Generally known as DoS attacks or negation of service, this attack involves the application of techniques that make the network unavailable.
- (3) Attack on integrity : This type of attack the attacker's goal is to inject false data on the network, keeping the network available, but traveling fictitious data.

3.6 Four Pillars of IoT

➡ **SPPU : May-18, Dec.-18, May-19, Dec.-19, March-20**

- Technology is rapidly changing our world through the Internet-of-Things. IoT is a network of physical objects communicating using embedded wireless capabilities. These objects include machines in factories (machine-to-machine, M2M), medical instruments in hospitals, utility meters in smart-grid systems, vehicles on highways (vehicle-to-vehicle, V2V) and food in refrigerators.
- Internet of Things is used by all type of industry. So it is necessary to classify the all vertical applications and identify common technologies that can be used across the board. To achieve new creative, disruptive applications, interconnecting, interrelating and synergized grand integration is necessary.
- One of the common characteristics of the IoT is that objects in a IoT world have to be instrumented, interconnected, before anything can be intelligently processed and used anywhere, anytime, anyway and anyhow, which are the 5A and 3I characteristics.
- To achieve such 5A and 3I capabilities, some common, horizontal, general-purpose technologies, standards and platforms, especially middleware platforms based on common data representations just like the three-tiered application server middleware, HTML language and HTTP protocol in the Internet / web world.
- It established to support various vertical applications cost effectively and new applications can be added to the platform unlimitedly.
- Fulfill the 5A concept, that means, it can be intelligently processed and used : For anything, Anywhere, Anytime, Anyway and Anyhow. From any time, any place connectivity for anyone, we will now have connectivity for anything.
- Fulfill the 3I concept, that means, is an object that is : Instrumented Interconnected Intelligent.
- Most of the vertical applications of IoT utilize common technologies from the networking level and middleware platform to the application level, such as standard wired and wireless networks, DBMS, security framework, web-based three-tiered middleware, multitenant PaaS, SOA interfaces and so on.

- Service Management Platforms (SMPs) allow for the essential connectivity management, intelligent rate-plan management and customer self-service capability that are today's fundamental prerequisites for providing a successful, managed M2M service.
- Telenor Objects aims to provide layered and horizontal architecture for connecting devices and application. Telenor Objects is a Norwegian service provider within Machine-to-Machine communication and constitutes one of the two Telenors strategic initiatives to catch the market for M2M, or connected objects.

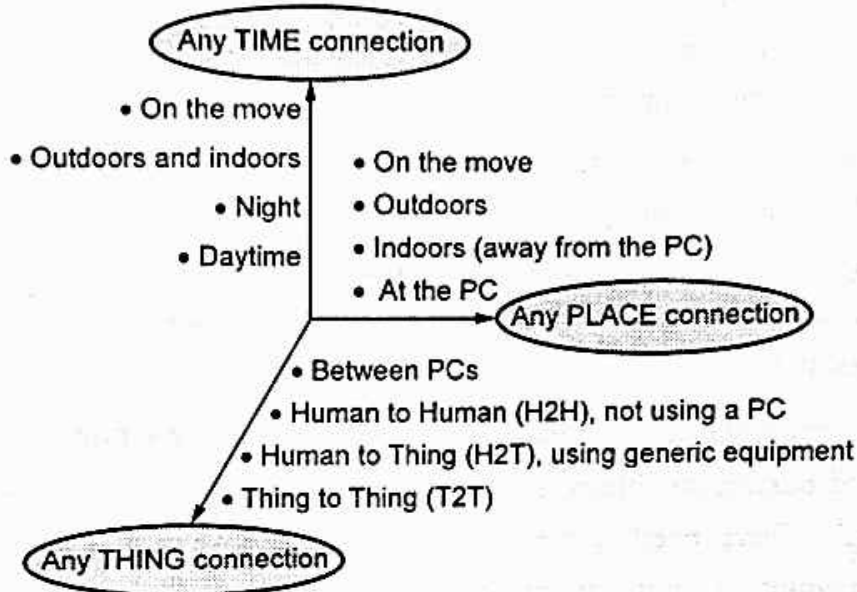


Fig. 3.6.1

- Telenor Objects delivers a managed service for connected objects. The aim is to enhance the extent and connectivity of network infrastructures that connect objects and provide real-time information from such distributed devices, sensors or other assets. The company supports and nurtures the internet of things concept-services that imply networked interconnection of everyday objects.
- Telenor Objects aims to provide a layered and horizontal architecture for connecting devices and applications. Telenor Objects was formed in July 2009, by researchers and developers in Telenor Norway and Telenor R&I. The two entities had individually been working on piloting managed M2M services since 2007, with an RFID focus in Telenor Norway and a focus on trace-and-track initiatives in Telenor R&I.
- In the vertical business model, the IoT device, the gateway (if used) and the cloud-based service are all provided and controlled by the same company. This approach has the advantage for the end-user that there are no compatibility issues to deal with among the various elements and a single point of contact to deal with if anything goes wrong. The disadvantages are that the end-user is entirely dependent on the vendor for improvements, enhancements, or upgrades to the offering.

- IoT home-security system that monitors an empty house for intruders, for instance, has the same hardware as one that monitors an elderly person's activity so that it can summon aide if the person falls or loses consciousness. But if someone wants a system that will do both, they are dependent on the system vendor to offer those features when dealing with a vertically defined business.
- Vertical business models can also result in users needing several different systems to achieve a spectrum of tasks, each with its own gateway and cloud operations. In a smart home, for instance, it is easy to end up with separate providers for security, HVAC and appliance systems. This complicates system management for the end-user.
- The motivation behind a horizontal model is to foster rapid growth and innovation in the industry by allowing multiple providers to work with a common framework. The idea is that by making the gateway and cloud resources something that can be assumed to be in place and have known and open functionality, innovators can concentrate their efforts on creating devices and services.
- The horizontal approach makes innovation easier and allows rapid proliferation of new applications and businesses, but it needs to gain considerable traction before it can pay off on its promises. There need to be a lot of open gateways in place before third-party developers can count on having an adequate market to serve.
- The key benefits of horizontal standard-based platforms will be faster and less costly application development and more highly functional, robust and secure applications.
- The six pillars of M2M are as follows :
 1. Remote monitoring is a generic term most often representing supervisory control, data acquisition and automation of industrial assets.
 2. RFID is a data-collection technology that uses electronic tags for storing data.
 3. A sensor network monitors physical or environmental conditions, with sensor nodes acting cooperatively to form / maintain the network.
 4. The term smart service refers to the process of networking equipment and monitoring it at a customer's site so that it can be maintained and serviced more effectively.
 5. Telematics to the integration of telecommunications and infomatics, but most often it refers to tracking, navigation and entertainment applications in vehicles.
 6. Telemetry is usually associated with industrial, medical- and wildlife-tracking applications that transmit small amounts of vehicles data.

- A four-pillar graphic is introduced for the broader IoT universe. The four pillars of IoT are M2M, RFID, WSNs and SCADA (supervisory control and data acquisition)
 1. RFID : Radio Frequency Identification, for identify or add tags to objects or people.
 2. WSN : Wireless Sensor Networks. Its specialty is to monitor environmental or physical conditions.
 3. M2M : Machine to Machine, machine communicating directly with other machines via wireless or wired network
 4. SCADA : Is a supervisory control and data acquisition specialized software used in industry, but it can be applied to small systems that consists of. HMI : Human Machine Interfaces. Could be a touch panel for user graphical information presentation.
- RFID uses radio waves to transfer data from an electronic tag attached to an object to a central system through a reader for the purpose of identifying and tracking the object.

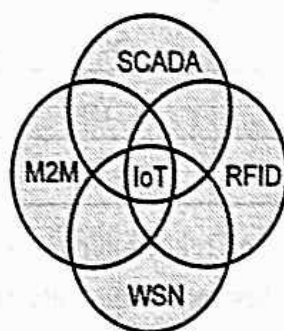


Fig. 3.6.2

- A WSN consists of spatially distributed autonomous sensors to monitor physical or environmental conditions.
- SCADA is an autonomous system based on closed-loop control theory or a smart system or a CPS data connects, monitors and controls equipment via network in a facility such as a plant or a building.
- Harbor Research defines smart systems as a new generation of systems architecture that provides real-time awareness based on inputs from machines, people, video streams, maps, new feeds, sensors and more that integrate people, processes and knowledge to enable collective awareness and decision making.
- People, information and technology are becoming more connected, distributed and pervasive enabling the convergence of physical and virtual worlds. Network awareness will include knowledge, people and things.

- These forces are informing a new trend we call "Smart Business." In its simplest form, Smart Business is a concept in which input from machines, people, video streams, maps, news-feeds, sensors and more is digitized and placed onto networks.
- These inputs are integrated into systems that connect people, processes and knowledge to enable collective awareness, creativity and better decision making. The foundation of Smart Business is made up of "Smart Systems" based on leveraging embedded computing and networking technology to deliver smart, remotely monitorable goods that will support entirely new modes of customer-device interaction and service delivery.
- Four Pillars of IoT and Their Relevance to Networks :

Four Pillars and Networks	Short- Range Wireless	Long- Range Wireless	Short- Range Wired	Long- Range Wired
RFID	Yes	Some	No	Some
WSN	Yes	Some	No	Some
M2M	Some	Yes	No	Some
SCADA	Some	Some	Yes	Yes

- The four pillars of IoT paradigms and related networks. IoT is the glue that fastens the four pillars through a common set of best practices, networking methodology and middleware platform.

3.6.1 M2M

- Machine to Machine (M2M) communication is the communication among the physical things which do not need human intervention.
- M2M communication is a form of data communication that involves one or more entities that do not necessarily require human interaction or intervention in the process of communication. M2M is also named as Machine Type Communication (MTC) in 3GPP.
- M2M communication could be carried over mobile networks (e.g. GSM-GPRS, CDMA EVDO networks). In the M2M communication, the role of mobile network is largely confined to serve as a transport network.
- M2M is only a subset of IoT. IoT is a more encompassing phenomenon because it also includes Human-to-Machine communication (H2M).
- Radio Frequency Identification (RFID), Location-Based Services (LBS), Lab-On-a-Chip (LOC), sensors, Augmented Reality (AR), robotics and vehicle telematics, which are some of the technology innovations that employ both M2M and H2M communications.

- **Reasons for shifting from M2M to IoT :**

1. It supports multiple application with multiple device.
2. It is information and service centric.
3. It supports open market place.
4. IoT uses Horizontal enabler approach.
5. It requires generic commodity devices.
6. Used in B2B and B2C.

- **Key features of M2M :**

1. Low Mobility : M2M Devices do not move and if moves only within a certain area.
2. Time Controlled : Data can be sent or receive only at certain pre-defined time periods.
3. Time Tolerant : Sometimes data transfer can be delayed.
4. Packet Switched : Network operator to provide packet switched service
5. Online small Data Transmissions : Devices frequently send or receive small amounts of data.
6. Low Power Consumption : To improve the ability of the system to efficiently service M2M applications.
7. Location Specific Trigger : Intending to trigger M2M device in a particular area e.g. wake up the device.

- **Six Pillars of M2M :**

- The six pillars of M2M are as follows :

1. Remote monitoring is a generic term most often representing supervisory control, data acquisition and automation of industrial assets.
2. RFID is a data-collection technology that uses electronic tags for storing data.
3. A sensor network monitors physical or environmental conditions, with sensor nodes acting cooperatively to form / maintain the network.
4. The term smart service refers to the process of networking equipment and monitoring it at a customer's site so that it can be maintained and serviced more effectively.
5. Telematics to the integration of telecommunications and infomatics, but most often it refers to tracking, navigation and entertainment applications in vehicles.
6. Telemetry is usually associated with industrial, medical and wildlife-tracking applications that transmit small amounts of vehicles data.

3.6.1.1 Architecture and Components of M2M

- Fig. 3.6.3 shows M2M architecture.

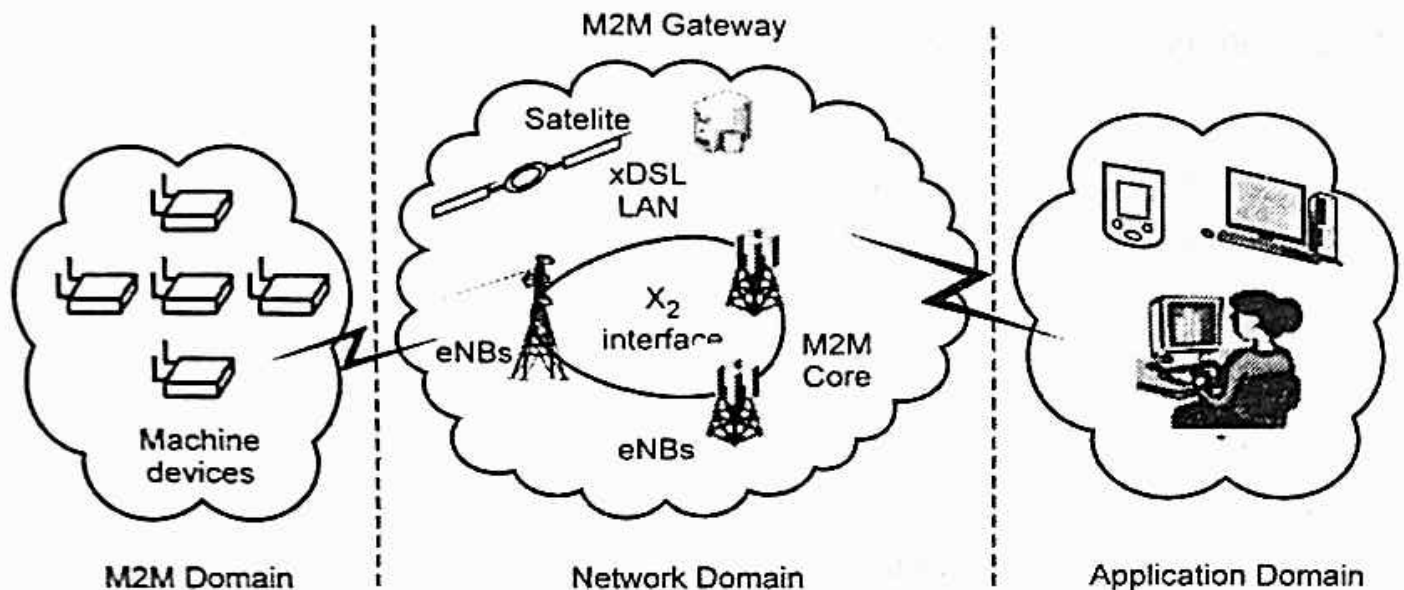


Fig. 3.6.3 : M2M architecture

- The system components of an M2M solution are as follows :

- 1. M2M device :** A device that runs application(s) using M2M capabilities and network domain functions. An M2M device is either connected straight to an access network or interfaced to M2M gateways via an M2M area network.
- 2. M2M area network :** A M2M area network provides connectivity between M2M devices and M2M gateways. Examples of M2M area networks include : Personal area network technologies such as IEEE 802.15, SRD, UWB, Zigbee, Bluetooth, etc or local networks such as PLC, M-BUS, Wireless M-BUS.
- 3. M2M gateways :** Equipments using M2M capabilities to ensure M2M devices interworking and interconnection to the network and application domain. The M2M gateway may also run M2M applications.
- 4. M2M applications server :** Applications that run the service logic and use service capabilities accessible via open interfaces.
- 5. M2M application :** The application component of the solution is a realization of the highly specific monitor and control process. The application is further integrated into the overall business process system of the enterprise.

- Fig. 3.6.4 shows generic M2M solution.

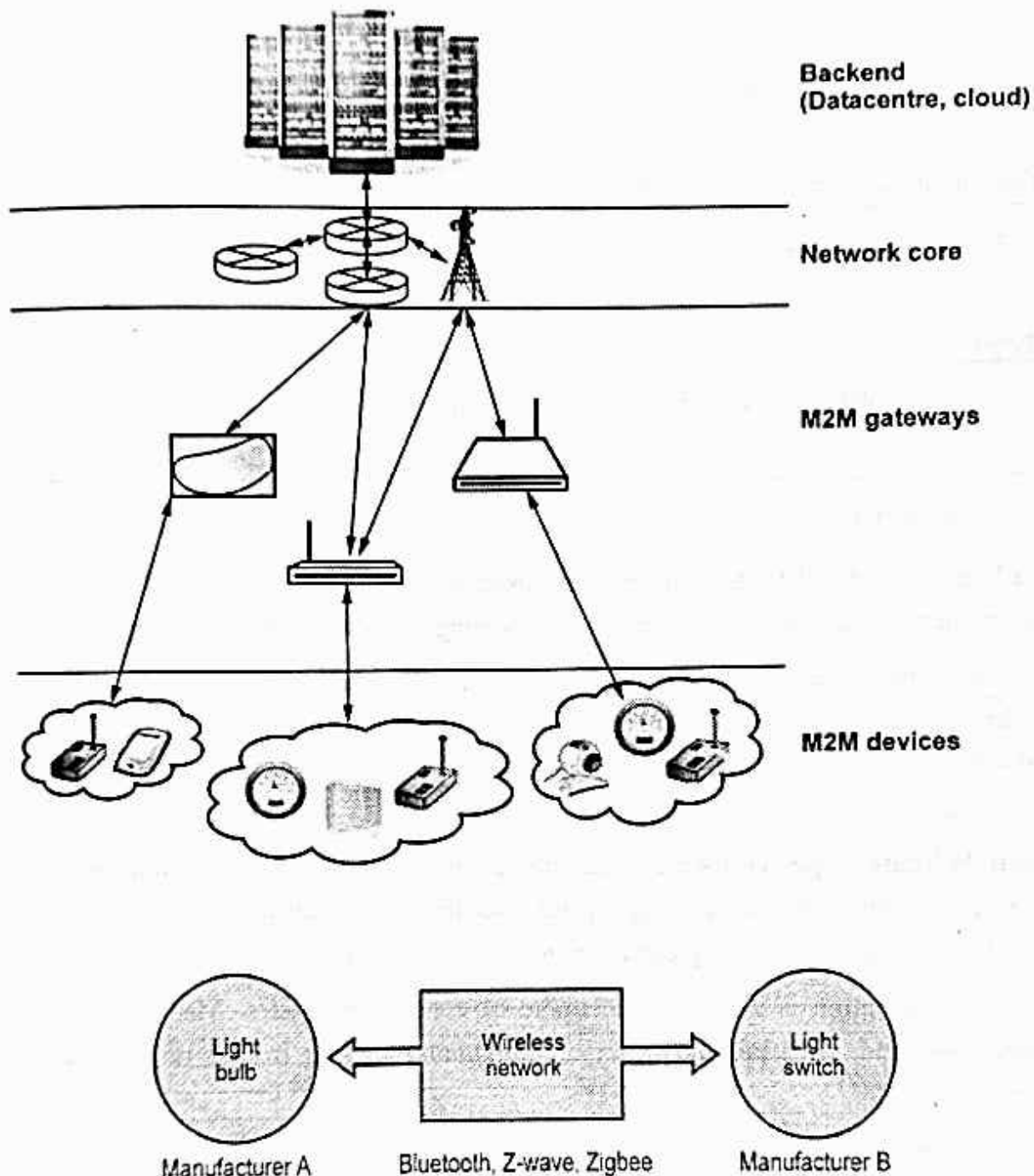


Fig. 3.6.4 : Generic M2M solution

- A number of sub-sets of users of M2M services can be identified : Consumers in the home, business users and facility managers, city governments, logistics businesses, energy providers and more.

3.6.1.2 Difference between M2M and IoT

Machine-to-Machine	Internet of Things
It support single application with single device.	It support multiple application with multiple device.
It is communication and device centric.	It is information and service centric.
It support closed business operations.	It support open market place.
M2M uses vertical system solution approach.	IoT uses horizontal enabler approach.
It requires specialized device solutions.	It requires generic commodity devices.
Used in B2B.	Used in B2B and B2C.

3.6.2 SCADA

- The Internet Protocol Suite is increasingly used on small devices with severe constraints on power, memory, and processing resources, creating constrained-node networks.
- A node where some of the characteristics that are otherwise pretty much taken for granted for Internet nodes at the time of writing are not attainable, often due to cost constraints and / or physical constraints on characteristics such as size, weight, and available power and energy.
- The tight limits on power, memory, and processing resources lead to hard upper bounds on state, code space, and processing cycles, making optimization of energy and network bandwidth usage a dominating consideration in all design requirements.
- Power consumption is a key characteristic of constrained nodes. Many IoT devices are battery powered, with lifetime battery requirements varying from a new months to 10+ years.
- IoT constrained nodes can be classified as follows :
 1. Devices that are very constrained in resources, may communicate infrequently to transmit a few bytes, and may have limited security and management capabilities.
 2. Devices with enough power and capacities to implement a stripped down IP stack or non-IP stack.
 3. Devices that are similar to generic PCs in terms of computing and power resources but have constrained networking capacities, such as bandwidth.

3.6.3 WSN

- WSN is more for sensing and information-collecting purposes. Other networks include Body Sensor Network (BSN), Visual or Video Sensor Network (VSN), vehicular sensor networks, underwater (acoustic) sensor networks, interplanetary sensor networks, fieldbus networks and others.
- The extended scope of WSN is the USN, or ubiquitous sensor network, a network of intelligent sensors that could one day become ubiquitous.
- The Wireless Sensor Network (WSN) developed at Linköping University in Sweden is based on the ZigBee specification and is an IoT solution. It provides nodes for monitoring temperature, relative humidity, carbon dioxide, VOC, PM and electricity consumption as well as for automatic control.
- A Wireless Sensor Network (WSN) is a network formed by a large number of sensor nodes where each node is equipped with some sensors to detect physical phenomena. In IoT, the sensor nodes and devices are interconnected to transmit useful measurement information via distributed sensor networks.
- Other networks are Body Sensor Network (BSN), Video Sensor Network (VSN), vehicular sensor network (V2V), interplanetary sensor network etc.
- VSN devices come with image sensors, adequate processing power and memory. They use wireless communication interfaces to collaborate and jointly solve tasks such as tracking persons within the network. In all applications, VSNs monitor a potentially large group of people and record sensitive image data which might contain identities of persons, their behavior, interaction patterns or personal preferences.
- A central idea of VSNs is to keep data processing local to reduce the amount of transmitted data. Fig. 3.6.5 shows VSN.

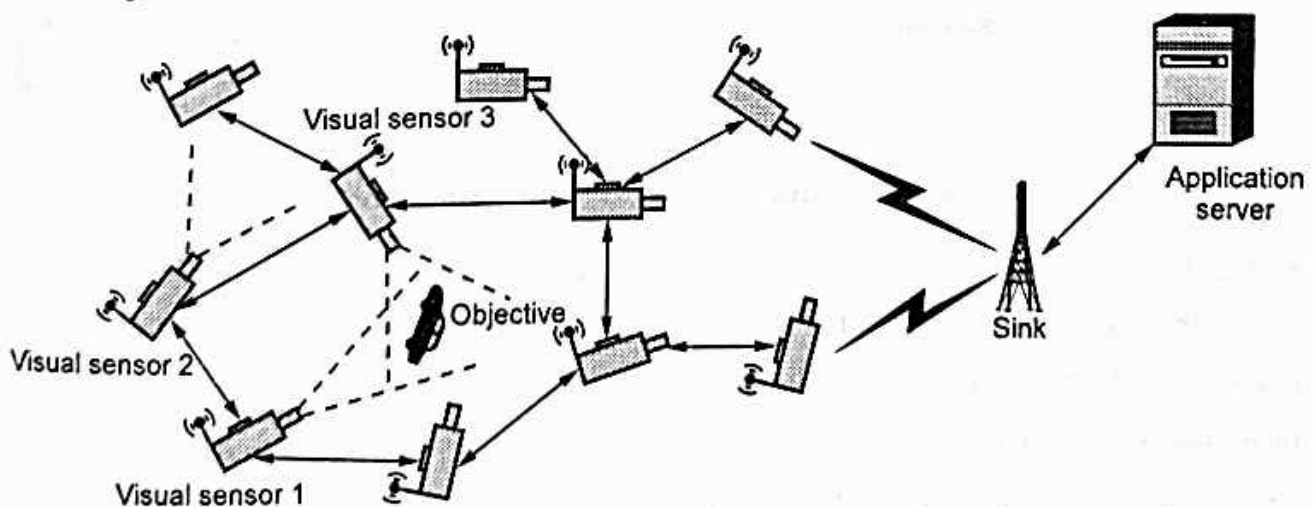


Fig. 3.6.5 : Example of wireless visual sensor networks

- A single VSN device has only a limited field of view but VSNs are typically designed to cover large areas. Therefore, multiple spatially distributed nodes are required. To avoid centralized control and data processing, VSNs use peer to peer communication for coordination, configuration, data exchange, handover of tracked objects or data fusion.
- To simplify deployment of spatially distributed VSNs, they rely no longer only on dedicated communication networks but make use of existing infrastructure which is not under full control of the VSN operators.
- Other VSN applications include environmental monitoring, smart homes and meeting rooms, entertainment and virtual reality as well as elderly care and assisted living.
- Body sensor network (BSN) is one of application of wearable computing device to enable wireless communication between several miniaturized body-sensor units and a single body central unit worn on the human body to transmit vital signs and motion readings to medical practitioners.
- Fig. 3.6.6 shows sources of body sensor network.

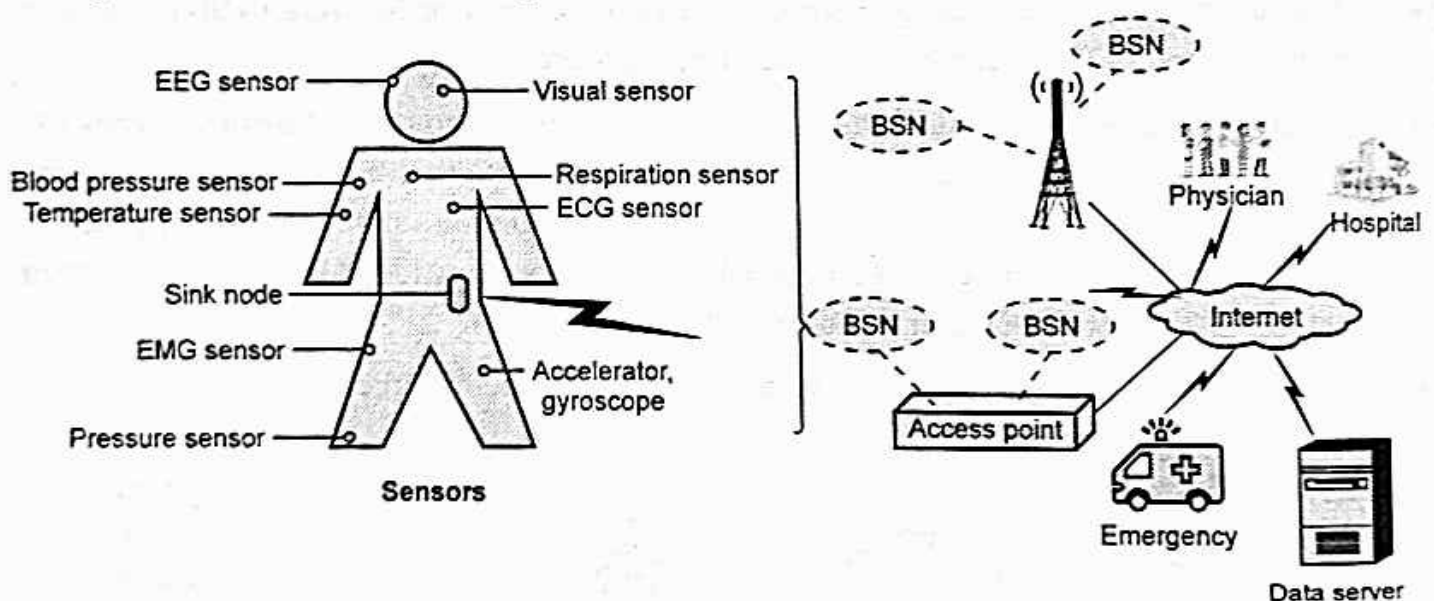


Fig. 3.6.6 : Source of body sensor network

- Sensor node electronics should be miniaturized, low power and detect medical signals such as electrocardiogram (ECG), photoplethysmogram (PPG), pulse rate, blood flow, pressure and temperature. The collected data from the control devices are then transferred to remote destinations in a wireless body area network for diagnostics.
- Most popular wireless technologies used for medical monitoring are ZigBee, WLAN, GSM, Bluetooth.

- Sensor nodes are designed to collect raw signals from a human body. A sensor node undertakes three functions : Detecting signal via a front end, digitizing / coding / controlling for a multi access communication and finally wireless transmission.
- In data acquisition and processing, the microcontroller maintains a power management scheme to control the distribution of the energy from battery in an optimized manner. The signal from a human body is usually weak and coupled with noise.
- The development of WSNs was motivated by military applications such as battlefield surveillance. Fig. 3.6.7 shows typical sensor network.

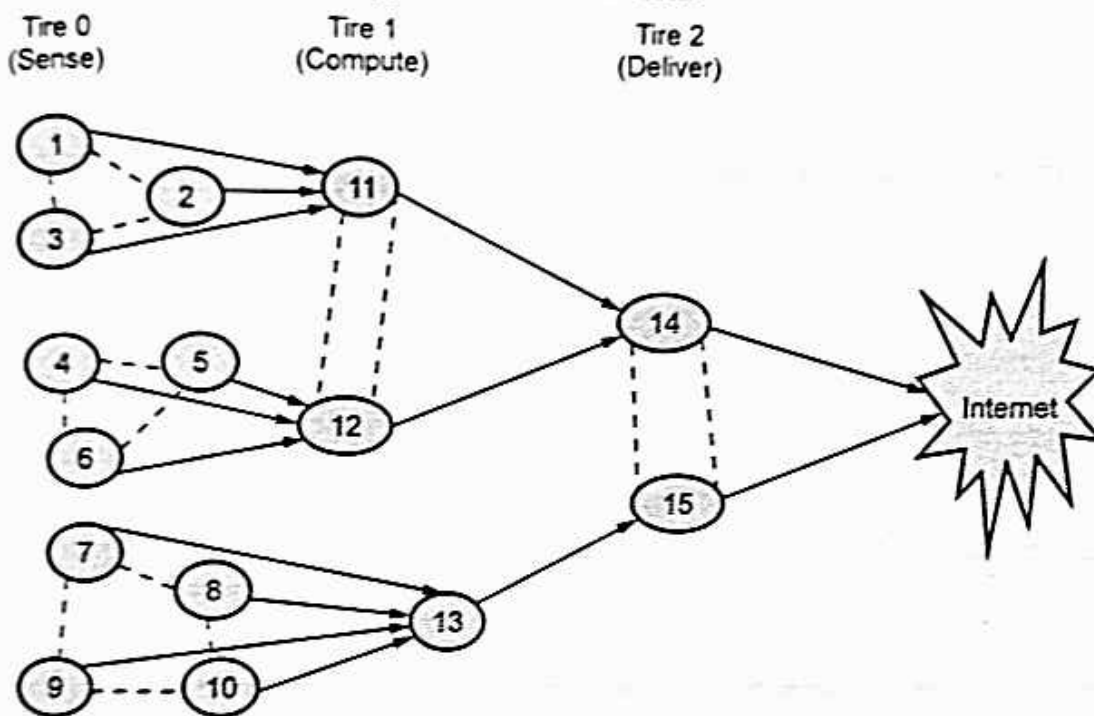


Fig. 3.6.7 : Sensor network architecture

- WSN is built of nodes, each node connected to one or more sensors. Each such sensor network node has typically several parts : A radio transceiver with an antenna, a microcontroller, an electronic circuit for interfacing with the sensors and an energy source, usually a battery or an embedded form of energy harvesting.
- Topology of sensor network may be star topology or mesh topology. Following are the components used by sensor networks :
 1. **Sensor node :** Sense target events, gather sensor readings, manipulate information, send them to gateway via radio link
 2. **Base station / sink :** Communicate with sensor nodes and user / operator
 3. **Operator / user :** Task manager, send query

- For reliable data transmission in a WSN, routing method is used. Routing protocols are distributed and reactive. The nodes in the system start looking for a route only when they have application data to transmit.
- Commonly used routing algorithm for WSN are Dynamic Source Routing (DSR) and Ad hoc On-Demand distance Vector (AODV).
- Energy is the scarcest resource of WSN nodes and it determines the lifetime of WSNs. WSNs are meant to be deployed in large numbers in various environments, including remote and hostile regions, with ad hoc communications as key. For this reason, algorithms and protocols need to address the following issues :
 1. Lifetime maximization
 2. Robustness and fault tolerance
 3. Self- configuration
- Some of the existing WSN platforms are summarized below. Most of the device designs are still in the research stage.
 1. Accsense, Inc
 2. Atlas
 3. Cortex Project
 4. Berkeley Motes
 5. Ambient Systems mesh networks (Netherlands)
 6. G-Node from SOWNet Technologies
 7. BTnode
 8. BEAN Project

□ Context-aware system based on WSN

- The term "context" is defined as "any information that can be used to characterize the situation of an entity". This definition is traditionally associated with the design of context-aware applications, where contexts are information that describes the situation of any entity relevant to the application.
- The entities can be people, places, or things. The applications are context consumers that receive information from context producers and may also produce their own context that they provide to other context consumers (e.g., other applications). By exchanging contexts, the applications can adapt their behaviors according to shared contexts and overall performance can be optimized as a result.

- Context can be shared between context producers and consumers in the network. Context for the IoT system is any shareable knowledge to represent situations or conditions from different parts of the system. The range of contexts can include, but not limited to, device context, network contexts, system contexts and environment contexts.
- The device context provides knowledge about local device conditions, such as the energy state, storage level and services provided by a node.
- The network context represents network wide situations and states, such as network topology, overall transmission capacity, or path qualities in the network.
- The system context represents status of the IoT system, such as the current executing tasks of an application or the state of the IoT system performance, which can be shared with the underlying network.
- The environment context provides knowledge for a network to understand the changes of its environmental properties or attributes, such as the occurrence of a fire incident or detection of hazardous objects.
- Mobile Sensor Networks (MSNs) are example of WSNs in which nodes can move under their own control. Mobile networked systems combine the most advanced concepts in perception, communication and control to create computational systems capable of interacting in meaningful ways with the physical environment, thus extending the individual capabilities of each network component and network user to encompass a much wider area and range of data.
- Difference between WSN and Body sensor network.

Wireless Sensor Network	Body sensor network
WSN contains large number of sensor nodes	BSN contains fewer sensor nodes
Cover the environment	Cover the human body
Fixed structure	Variable structure
Low level security	High security
Accessible power supply	Inaccessible power source
Low context awareness	High context awareness

3.6.4 RFID

- Radio Frequency Identification (RFID) is a very simple and cost-effective way of item identification. RFID systems can be seen as a next-generation technology for bar-codes. RFID devices are wireless microchips used for tagging objects for automated identification.

- An RFID tag is a simplified, low-cost, disposable contactless smartcard. RFID tags include a chip that stores a static number (ID) and attributes of the tagged object and an antenna that enables the chip to transmit the store number to a reader.
- Tags are characterized by a unique identifier and are applied to objects. Readers trigger the tag transmission by generating an appropriate signal, which represents a query for the possible presence of tags in the surrounding area and for the reception of their IDs. Fig. 3.6.8 shows basic RFID system.

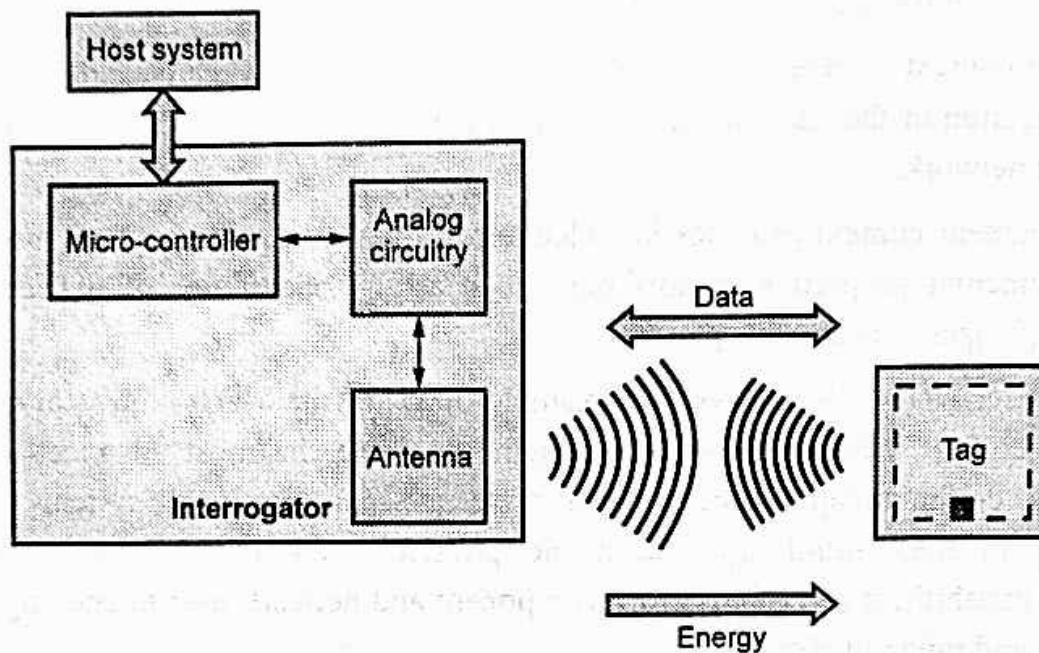


Fig. 3.6.8 : Basic RFID system

- RFID systems consist of a reading device called a reader and one or more tags. The reader is a powerful device with ample memory and computational resources.
- Passive tags have limited computational capacity, no ability to sense the channel, detect collisions and communicate with each other. They respond only at reader commands.
- Semi-passive tags have an on-board power source that can be used to energize their microchip. Active tags can sense the channel and detect collisions.
- Accordingly, RFID systems can be used to monitor objects in real-time, without the need of being in line-of-sight; this allows for mapping the real world into the virtual world.
- An RFID system involves hardware known as readers and tags, as well as RFID software or RFID middleware. Readers can also be mobile / hand-held.
- RFID systems operate in the Industry, Scientific and Medical (ISM) frequency band that ranges from 100 kHz to 5.8 GHz.

❑ RFID Frequencies Range :

- RFID tags may operate in different frequencies with respect to their initial designs. Different frequencies and their respective devices offer different advantages.

RFID	Key Applications	Standard
125 kHz (Low Frequency)	Inexpensive passive RFID tags for identifying animals	ISO 18000-2
13.56 MHz (high Frequency)	Inexpensive passive RFID tags for identifying objects. Example : library book identification, clothes identification	ISO 14443
400 MHz	For remote control for vehicle centre locking systems	ISO 18000-7
868 MHz, 915 MHz and 922 MHz (UHF)	For active and passive RFID tags for logistics in Europe, US and Australia respectively.	
2.45 GHz (Microwave)	An ISM band, used for active and passive RFID tags. Used for temperature sensor or GPS collection	ISO 18000-5

- For example, RFID systems can track items in real-time with the help of Global Positioning System (GPS), which provides essential information about the objects like physical location or status. Sensor technologies are also emerging in this regard to collect and transmit the physical status or data from the objects or things.
- Reader functions :
 1. Remotely power tags
 2. Establish a bidirectional data link
 3. Inventory tags, filter results
 4. Communicate with networked server(s)
 5. Can read 100-300 tags per second

❑ RFID Anti-Collision Protocols

- Collision due to simultaneous tag responses is one of the key issues in RFID systems. Tag collision results in wastage of bandwidth, energy and increases identification delays.

- RFID readers must use an anti-collision protocol to minimize collisions and hence help reduce identification delays. Fig. 3.6.9 shows tag collision problem.

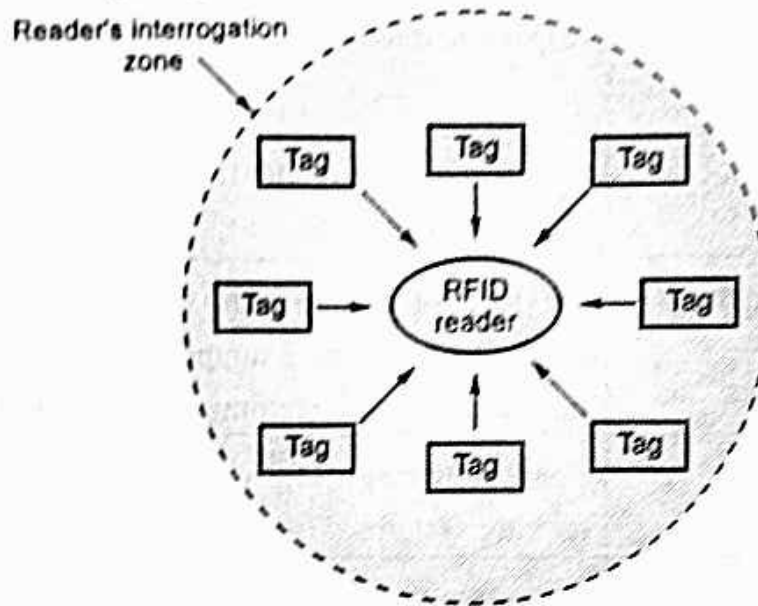


Fig. 3.6.9 : Tag collision problem

- RFID anti-collision protocols are often categorized as Aloha based protocols and tree based protocols. In pure Aloha based RFID systems, a tag responds with its ID randomly after being energized by a reader.
- In Slotted Aloha (SA) based RFID systems, tags transmit their ID in synchronous time slots. The collision occurs at slots boundary only, hence there are no partial collisions

☐ RFID advantages over bar-codes

1. No line of sight required for reading
2. Multiple items can be read with a single scan
3. Each tag can carry a lot of data (read / write)
4. Individual items identified and not just the category
5. Passive tags have a virtually unlimited lifetime
6. Active tags can be read from great distances
7. Can be combined with barcode technology

☐ RFID Middleware

- RFID middleware needs to allow users to configure, deploy and issue commands directly to readers through a common interface. For instance users should be able to tell a reader when to "turn off" if needed. Fig. 3.6.10 shows RFID middleware.

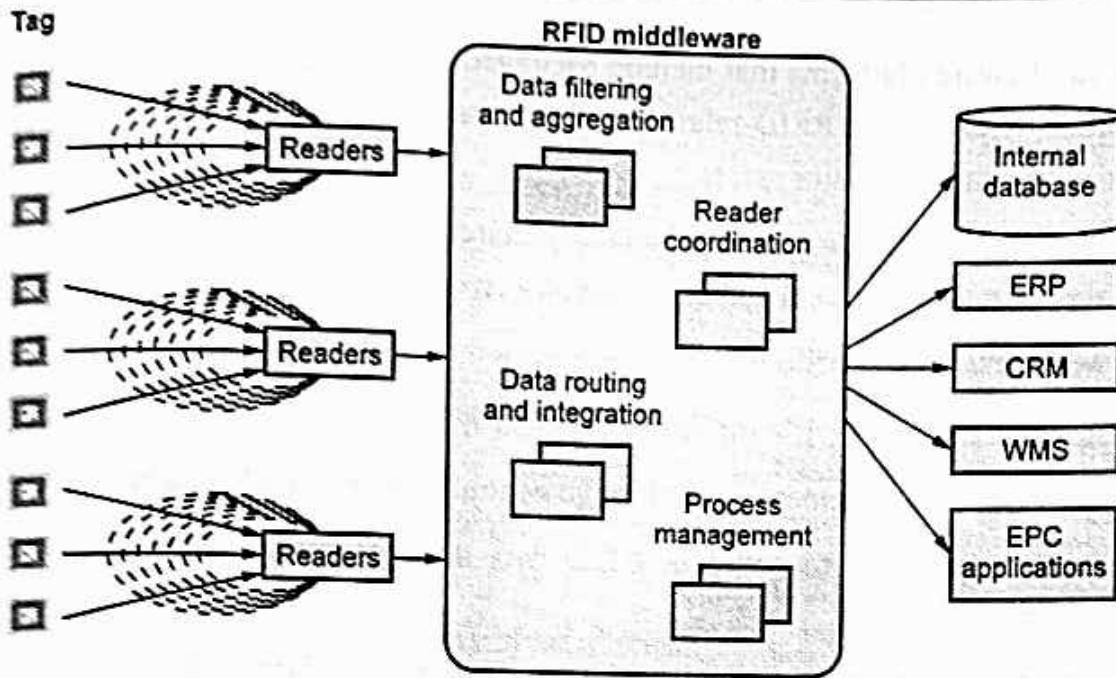


Fig. 3.6.10 : RFID middleware

- After RFID middleware captures EPC data from readers, it must be able to intelligently filter and route the data to the appropriate destinations.
- Look for middleware that includes both low-level logic and more complex algorithms. Comprehensive solutions also offer tools for aggregating and managing EPC data in either a federated or central data source.
- RFID middleware solutions need to provide the messaging, routing and connectivity features required to reliably integrate RFID data into existing SCM, ERP, WMS, or CRM systems. Ideally through a Services-Oriented Architecture (SOA).
- A services-oriented architecture is essentially a collection of services. These services communicate with each other. The communication can involve either simple data exchange or two or more services coordinating some activity, such as order placement or inventory control.
- Middleware needs to provide a library of adapters to popular WMS and SCM applications (e.g., SAP or Oracle E Business Suite). Application Programming Interfaces (APIs) and adapters for using standard technologies like JMS, XML and SOAP to integrate with other third-party applications.
- RFID middleware must provide :
 1. B2B integration features like partner profile management
 2. Support for B2B transport protocols
 3. Integration with a partner's data over communications such as EDI, Web-based systems like AS2, or a well engineered system specifically for EPC data RFID Middleware

- RFID middleware platforms that include packaged routing logic, product data schemas and integration with typical RFID-related applications and processes like shipping, receiving and asset tracking are major assets.
- RFID middleware platforms must include features for dynamically balancing processing loads across multiple servers and automatically rerouting data

☐ **RFID Middleware benefits**

- a. Minimized network traffic through intelligent filtering
- b. Lower reader-management costs through centrally coordinated readers
- c. Immediate visibility to pertinent RFID data through routing, filtering and track-and-trace tools
- d. Minimized on-going integration costs through standard APIs and prepackaged application integration tools
- e. Well-architected RFID middleware can enable more strategic opportunities that go way beyond these initial, rather obvious benefits

☐ **Industrial plants**

- Smart environments also help in improving the automation in industrial plants with a massive deployment of RFID tags associated to the production parts. In a generic scenario, as production parts reach the processing point, the tag is read by the RFID reader.
- An event is generated by the reader with all the necessary data, such as the RFID number and stored on the network. The machine/robot gets notified by this event and picks up the production part.
- By matching data from the enterprise system and the RFID tag, it knows how to further process the part. In parallel, a wireless sensor mounted on the machine monitors the vibration and if it exceeds a specific threshold an event is raised to immediately stop the process.
- Once such an emergency event is propagated, devices that consume it react accordingly. The robot receives the emergency shut-down event and immediately stops its operation. The plant manager also immediately sees the status of the so called Enterprise Resource Planning (ERP) orders, the production progress, the device status, as well as a global view on all the elements and the possible side effects of a production line delay due to shop-floor device malfunctions.

❑ Weaknesses of RFID

1. Lack of industry and application standards
2. High cost per unit and high RFID system integration costs
3. Weak market understanding of the benefits of RFID technology

Review Questions

1. Explain WSN (the internet transducers) pillar of IoT.

SPPU : May-18, Dec.-19, Marks 4

2. What is SCADA ? What are the different blocks of SCADA.

SPPU : May-18, March-20, Marks 5

3. Explain M2M (the internet of devices) pillar of IoT.

SPPU : Dec.-18, Marks 4

4. Explain SCADA middleware standards in brief.

SPPU : Dec.-18, Marks 4

5. What is SCADA ? Explain the components of it.

SPPU : May-19, Marks 5

📖 3.7 Home Automation using IoT

- Home automation is the automatic control of electronic devices in your home. These devices are connected to the Internet, which allows them to be controlled remotely.
- Interconnected devices enable to intelligently monitor and control smart homes in a future Internet of Things.
- Energy saving applications, for example, control indoor climate and electricity usage by employing context information to switch off appliances (e.g., lights, computers), reduce room temperature, close windows, or stop warm water circulation.
- Home automation works on three levels :
 1. **Monitoring** : Monitoring means that users can check in on their devices remotely through an app. For example, someone could view their live feed from a smart security camera.
 2. **Control** : Control means that the user can control these devices remotely, like planning a security camera to see more of a living space.
 3. **Automation** : Finally, automation means setting up devices to trigger one another, like having a smart siren go off whenever an armed security camera detects motion.

3.7.1 Smart Lighting

- Smart control the lights with automation signal system to save energy. Smart, connected lighting is the next - generation energy - efficient LED products with additional sensors to sense things such as occupancy and temperature.

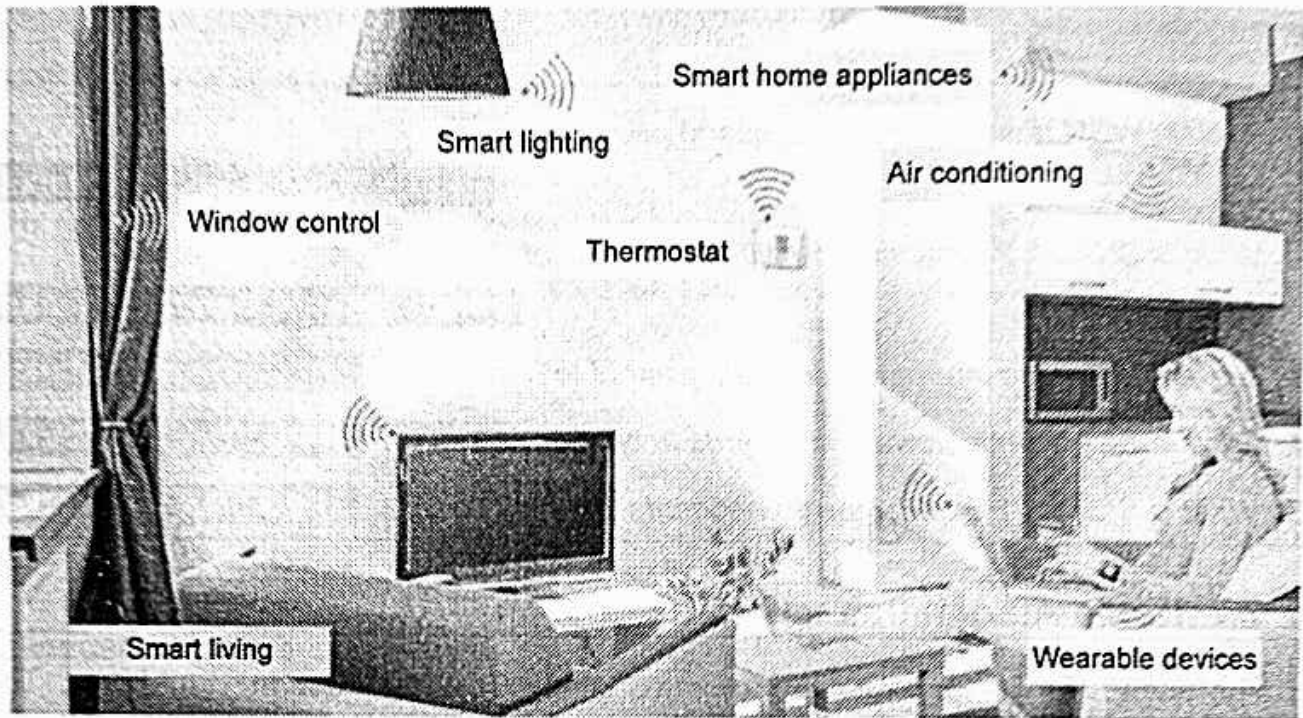


Fig. 3.7.1 : Smart home

- In automatic light control system, Light Dependent Resistor (LDR) sensor is used to detect bright /medium /dim /dark conditions.
- It is simple enough to envision the addition of sensors and communications to create that initial concept of smarter, more adaptive lighting. If people are present, turn the lights on; if not, turn them off. Or use your smart phone to connect to the lighting system and tune it to the desired brightness level or to a particular color.
- Smart lighting is considered the one of the main solutions for energy reduction by means of controlling lighting level according to desired need with minimum energy consumption.
- Smart - lighting systems utilize motion and light sensors for performing the control algorithms.
- The system uses motion and light sensors for detecting the surrounding environment. There are lamps controlled with the specific lighting level in order to supply the adequate amount of lighting required without affecting the user visibility.

- Certainly the required lighting level is strongly dependent on the weather conditions. In clear weather at night might require more luminance than cloudy one, due to the reflection from the clouds.
- While during mist and foggy weathers require the highest possible lighting level, as the visibility reaches its lowest. On snowy weather it might require an intermediate level between clear and foggy.
- During night it requires high lighting levels, while at day it needs just fade level to provide guidance or turn off if the weather is clear. The lighting concentration in the yard is affected by the above conditions.

3.7.2 Smart Appliances

- The role and scope of smart appliances in the home (Washer, dryer, refrigerator, dishwasher, fridge, freezer, air conditioner, vacuum cleaner and so on) is on the increase with the market being estimated to have a year on year compound growth of slightly over 15 %.
- Connecting everyday objects to the internet is an essential element of the IoT. Some appliance suppliers use a low power wireless network to communicate over such as bluetooth, whilst others utilise the existing higher powered Wi-Fi network used for a tablet or computer wireless connectivity. Once a network is in place objects can populate the home environment and communicate with the user and each other.
- The ability of an object to respond to remote commands and change its behaviour makes it an active device, such as the new Hive heating thermostat or a Sky+ box.
- Where the remote object has no ability to respond to remote control requests then it is considered passive as with some fixed cameras, microphones or temperature sensors.
- IR Sensor : It will be activated in the automated mode to detect person entering or coming out of the room and set a counter based on that. If the counter show there is a person inside it will light up the room automatically and turn on the AC depending upon the temperature reading.
- Sensors provide data about motion, occupancy, glass breakage, door and window openings, water leaks, light intensity, temperature, energy consumption, camera, and even appliance plug insertion or removal.
- Controllers turn power on and off or adjust settings on appliances, furnaces, air conditioners, space heaters, fans, pool pumps, water heaters, lighting, home theatres, music, motorized blinds, door locks, and plug loads.

- To be deemed intelligent, an appliance's sensors and controllers should use internet protocol communication.
- Smart refrigerators can keep track of the items stored and send updates to the users when an item is low on stock.
- Smart TV allow user to search video and movies from the Internet on a local storage drive, search TV schedule, fetch news and other things from the Internet.
- OpenRemote is the professional open source middle-ware for an Internet of Things. Integrate any device or protocol, and design your own user interface and automation. Use our online designer, sync to the controller, and control with this app.
- OpenRemote is a state of the art open source software platform for building control and automation.
- OpenRemote allows for designing a fully customizable building and home control solution without the need to actually write code.

3.7.3 Intrusion Detection

- Intrusion Detection System (IDS) includes both hardware and software mechanisms and IDS is responsible for identifying malicious activities by monitoring network environment and system.
- The purpose of home intrusion detection system is to detect intrusions using sensors and raise alerts, if necessary.
- With the help of Light dependent resistor and PIR motion sensor, it detect the motions in the room. If a motion is detected, system capture the image with the help of a webCam and store locally. Now the alerts are sent to the user with the captured image.
- Fig. 3.7.2 shows block diagram of intrusion detection.

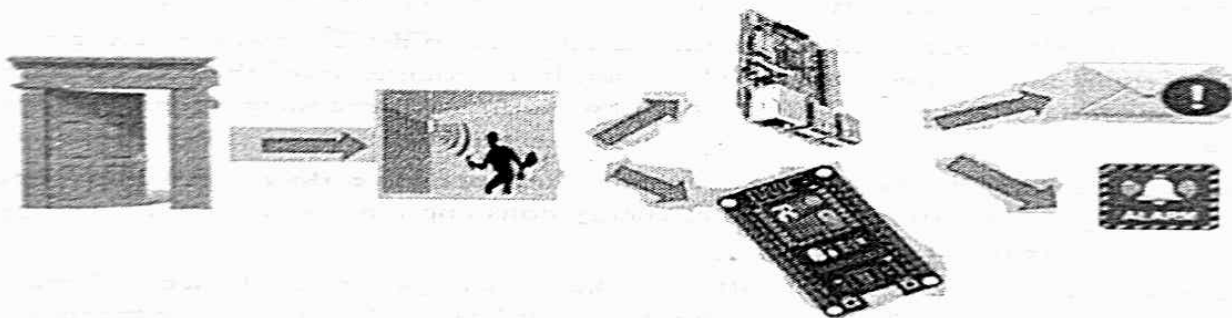


Fig. 3.7.2 : Block diagram of Intrusion detection

- To detect any form of intrusion in restricted areas and report it immediately, following concept is used.
 1. A PIR sensor is required to detect the presence of any human being in the room.
 2. An RFID is required to validate the presence of the person in the room by tallying his identity with those in the database.
 3. A camera is required to click the picture of the room and send it via email as an alarm.
 4. An internet connection is required to register all these movements on a website so that it can be accessed from any place and any device.
- The different input / output devices are controlled using TCP / IP over the IEEE 802.11 standard protocol. Data being gathered from sensors, such as PIR sensors, temperature sensors, IR transmitter and receiver is being processed on micro - controller as a server.
- Passive Infrared Sensor (PIR) Sensor : PIR sensor is an electronic sensing device that senses infrared (IR) light emitted from entities in its field of view and used to detect motion in its range. It is activated only in the security mode to detect any unwanted motion at the entrance. If any unwanted movement is detected then it will signal the microcontroller to take necessary steps.
- Alarm : It will only be activated in the security mode when some intruder is detected by the PIR motion sensor.
- Cloud controlled intrusion detection is possible by using location aware services. Here geo - location of each node is independently detected and stored in the cloud.
- Some intrusion detection system uses UPnP technology. It is based on image processing to recognize the intrusion.

3.7.4 Smoke for Gas Detection

- Smoke or gas detector sensor which detects the smoke and turns on the buzzer alarm and all these are update on the web page.
- MQ2 is a semiconductor type sensor, which can appropriately sense the presence of smoke, LPG, methane, butane, propane and other hydrocarbon.
- When it comes in contact with the gas to be monitored, the electrical resistance of the sensor decreases; enabling the microcontroller to respond to the situation.
- When it detects the concentration of combustible gas in the air it outputs its reading as an analog voltage. The sensor can measure concentrations of flammable gas of 300 to 10,000 ppm. The sensor can operate at temperatures from - 20 to 50 °C and consumes less than 150 mA at 5 V.

- The MQ-2 smoke sensor reports smoke by the voltage level as output. The more smoke is there, the greater the voltage output. The MQ-2 also has a built-in potentiometer to adjust the sensitivity to smoke.
- By adjusting the potentiometer, one can change how sensitive it is to smoke, so there is a form of calibrating it to adjust how much voltage it will give in relation to the smoke it is exposed to.